

#1 · Patient Care

What is informed consent?

Answer

The patient has been told, in language they understand, what a procedure involves, its risks and benefits, and what alternatives exist, and the patient or their legal representative agrees before the procedure happens.

#2 · Patient Care

What is oral consent?

Answer

Spoken agreement to a procedure, commonly used for simpler diagnostic imaging exams.

#3 · Patient Care

What is implied consent?

Answer

Consent inferred from the patient's behavior, such as rolling onto the table and holding still for positioning, or consent presumed under the emergency doctrine when an unconscious patient needs urgent treatment.

#4 · Patient Care

What is the emergency doctrine?

Answer

A legal principle allowing treatment of an unconscious patient without consent when delay in treatment would cause serious harm.

#5 · Patient Care

When is written consent generally required for imaging?

Answer

For invasive procedures involving contrast media or intervention, such as an arthrogram or an angiographic study.

#6 · Patient Care

What does HIPAA stand for and what does it protect?

Answer

The Health Insurance Portability and Accountability Act; it restricts who can access, discuss, or transmit a patient's protected health information.

#7 · Patient Care

What counts as protected health information (PHI) under HIPAA?

Answer

Anything that could identify the patient combined with their health information, such as name, date of birth, diagnosis, or images.

#8 · Patient Care

Who may access a patient's PHI under HIPAA?

Answer

Only staff with a legitimate need to know for treatment, payment, or health care operations.

#9 · Patient Care

What is a DNR order?

Answer

Do Not Resuscitate, a physician's written order stating that a patient should not receive CPR or advanced life support if their heart or breathing stops.

#10 · Patient Care

What is an advance directive?

Answer

A document a patient completes while competent that states their wishes for future medical care.

#11 · Patient Care

What is a living will?

Answer

A document that specifies what treatments a patient does or does not want if they become unable to communicate.

#12 · Patient Care

What is a health care proxy?

Answer

Also called a durable power of attorney for health care, it names another person to make medical decisions on the patient's behalf.

#13 · Patient Care

Can a patient be enrolled in a research study without separate consent?

Answer

No, patients cannot be enrolled in research without separate, specific informed consent for that research.

#14 · Patient Care

Define battery in the legal sense used on the ARRT exam.

Answer

Intentional, unauthorized physical contact with another person; performing an exam on a patient who has refused it can constitute battery even if done with good intentions.

#15 · Patient Care

Define negligence.

Answer

The failure to act as a reasonably prudent radiographer would act under similar circumstances, resulting in harm, such as forgetting to raise a side rail on a sedated patient who then falls.

#16 · Patient Care

Define malpractice.

Answer

Professional negligence, meaning negligence committed by someone acting in a professional capacity, such as a radiographer.

#17 · Patient Care

Define beneficence.

Answer

The ethical principle of acting in the patient's best interest, doing good rather than harm.

#18 · Patient Care

What does respondeat superior mean?

Answer

Latin for "let the master answer," this legal doctrine holds an employer legally responsible for the negligent acts of an employee performed within the scope of their job duties.

#19 · Patient Care

What does res ipsa loquitur mean?

Answer

"The thing speaks for itself," this doctrine applies when an injury is of a type that would not normally occur without negligence, such as a patient dropped from a table, and shifts the burden of proof toward the defendant.

#20 · Patient Care

What is the standard for correct patient identification before an exam?

Answer

Use at least two identifiers, such as the patient's full name and date of birth, and compare them against the order and, in an inpatient setting, the wristband.

#21 · Patient Care

Besides identity, what else should be checked before proceeding with an exam?

Answer

Compare the physician's order to the clinical indication to catch mismatches, such as an order for a left knee when the indication describes right knee pain.

#22 · Patient Care

Are positioning aids like sponges and sandbags considered restraints?

Answer

No. They are used routinely without special orders because the patient can remove them and they are not intended to restrict the patient's freedom for behavioral reasons.

#23 · Patient Care

When are restraints permitted and what governs their use?

Answer

Restraints require a physician's order except in emergencies, must be used only when necessary for the patient's safety, applied correctly to avoid injury, and monitored regularly.

#24 · Patient Care

What is the exposure indicator and why must it be recorded accurately?

Answer

A number reflecting how much radiation actually reached the detector; falsifying it to hide a repeat or an overexposure is a serious violation.

#25 · Patient Care

When does post-processing of a digital image become fraudulent?

Answer

When cropping or masking anatomy is done specifically to hide a positioning error or make an unacceptable image appear acceptable, rather than to adjust brightness or contrast within reasonable limits.

#26 · Patient Care

What must be documented in a patient's record regarding events during imaging?

Answer

Any change to an order, any medical event such as a fall or a reaction, and any repeated exposure, since the record is a legal document.

#27 · Patient Care

What is the difference between the ARRT Code of Ethics and the Rules of Ethics?

Answer

The Code of Ethics is an aspirational guide to professional behavior, while the Rules of Ethics are mandatory standards that, if violated, can result in disciplinary action including reprimand, probation, or revocation of certification.

#28 · Patient Care

What are the three channels of patient communication?

Answer

Verbal, written, and nonverbal communication.

#29 · Patient Care

Why does nonverbal communication matter so much with an anxious patient?

Answer

Eye contact, facial expression, tone of voice, and appropriate touch often carry more weight than the words spoken, and help establish trust before the procedure is explained.

#30 · Patient Care

Who should interpret for a patient with a language barrier?

Answer

A qualified medical interpreter, not a family member, whenever possible, since family members may filter information, lack medical vocabulary, or compromise the patient's privacy.

#31 · Patient Care

How should a radiographer adapt when imaging a hearing-impaired patient?

Answer

Face the patient directly so they can read lips.

#32 · Patient Care

What is the teach-back method?

Answer

A strategy to check understanding by asking the patient to repeat back what they heard about the procedure.

#33 · Patient Care

What are common pre-exam instructions for a contrast study?

Answer

Dietary restrictions such as NPO for several hours before an upper GI series, and medication adjustments such as holding metformin around iodinated contrast administration.

#34 · Patient Care

What should post-exam instructions after a contrast study cover?

Answer

Increasing fluid intake to help clear contrast media, watching for delayed reactions, and any discharge instructions specific to the procedure.

#35 · Patient Care

What are the three principles of good body mechanics?

Answer

Balance (wide base of support with feet shoulder-width apart), alignment (straight back, lifting with the legs), and controlled movement (moving smoothly and getting help when needed).

#36 · Patient Care

What is a Hoyer lift used for?

Answer

A mechanical, sling-based lift used for patients who cannot assist at all, such as a paralyzed or severely weak patient.

#37 · Patient Care

What is a transfer board?

Answer

A smooth, rigid board bridged between the stretcher and the table that a patient slides across during transfer.

#38 · Patient Care

What is a gait belt used for?

Answer

A fabric belt placed around a patient's waist that gives the technologist a secure handhold when assisting an ambulatory but unsteady patient to stand or walk.

#39 · Patient Care

How should an IV pump be positioned during imaging, and why?

Answer

Above the level of the heart to prevent backflow; it should be repositioned, not disconnected, when possible, and tubing must never be pulled taut.

#40 · Patient Care

How should a Foley catheter drainage bag be handled during imaging?

Answer

Always kept below the level of the bladder to prevent urine reflux and infection; the bag should never be set on the floor or hung on a moving part of the table.

#41 · Patient Care

What special consideration applies to a tracheostomy patient during imaging?

Answer

These patients cannot speak normally and may need alternative communication methods, and suction equipment should be accessible nearby.

#42 · Patient Care

What is the normal adult resting pulse rate range?

Answer

About 60 to 100 beats per minute.

#43 · Patient Care

What is the normal adult resting respiratory rate range?

Answer

About 12 to 20 breaths per minute.

#44 · Patient Care

What is a normal adult blood pressure reading?

Answer

Around 120/80 mm Hg.

#45 · Patient Care

What is a normal oral body temperature?

Answer

About 98.6 degrees Fahrenheit, or 37 degrees Celsius.

#46 · Patient Care

Name three fall prevention measures for imaging patients.

Answer

Locking wheelchair and stretcher wheels, raising side rails when appropriate, and never leaving a confused or sedated patient unattended on an imaging table.

#47 · Patient Care

What supplies commonly contain latex and can trigger an allergic reaction?

Answer

Some gloves, tourniquets, and catheter balloons.

#48 · Patient Care

What are the first two steps when a patient goes into cardiac or respiratory arrest?

Answer

Call for help or activate the emergency response system, then begin CPR if you are trained and certified.

#49 · Patient Care

What should you do if a patient has a seizure during imaging?

Answer

Protect the patient from injury by clearing the area and cushioning the head, rather than restraining the limbs or placing anything in the mouth.

#50 · Patient Care

What are the symptoms of hypoglycemia?

Answer

Confusion, sweating, and shakiness, from low blood sugar.

#51 · Patient Care

What are the symptoms of hyperglycemia?

Answer

Excessive thirst and fruity-smelling breath; hyperglycemia develops more slowly than hypoglycemia.

#52 · Patient Care

How should a suspected spine injury from a fall be handled?

Answer

Stabilize the patient, avoid moving them unnecessarily, and immediately notify the radiologist and nursing staff.

#53 · Patient Care

What are the six links in the chain of infection?

Answer

Pathogen, reservoir, portal of exit, mode of transmission, portal of entry, and susceptible host.

#54 · Patient Care

What are the two types of direct transmission?

Answer

Droplet spread (coughing or sneezing onto someone nearby) and direct contact (touching an infected person or their bodily fluids).

#55 · Patient Care

What are the three types of indirect transmission?

Answer

Airborne spread (small particles traveling on air currents, such as tuberculosis), vehicle-borne spread (a contaminated object like a doorknob or an X-ray table), and vector-borne spread (carried by an insect or animal).

#56 · Patient Care

What is the difference between disinfection and sterilization?

Answer

Disinfection reduces the number of pathogens on a surface but does not necessarily kill all spores, while sterilization destroys all microorganisms including spores.

#57 · Patient Care

What is medical asepsis?

Answer

Also called clean technique, it reduces the number of organisms present and their spread, including practices like hand hygiene and wearing clean gloves.

#58 · Patient Care

What is sterile technique?

Answer

Also called surgical asepsis, it eliminates all organisms from a field and is used for invasive procedures such as arthrograms.

#59 · Patient Care

How long should hands be washed with soap and water under Standard Precautions?

Answer

At least 20 seconds, before and after every patient contact.

#60 · Patient Care

Do Standard Precautions apply only to patients known to be infectious?

Answer

No, they apply to every patient, every time, regardless of known infection status, because you cannot always know who is infectious.

#61 · Patient Care

What is required for safe injection practices?

Answer

A new sterile needle and syringe for every injection, and never reusing a syringe on more than one patient, even to draw from a shared multi-dose vial.

#62 · Patient Care

What PPE is used for contact precautions and for which organisms?

Answer

Gown and gloves, used for organisms spread by touch, such as MRSA or C. difficile.

#63 · Patient Care

What PPE is used for droplet precautions and for which organisms?

Answer

A surgical mask, and eye protection if splash is likely, used for organisms spread by large respiratory droplets over short distances, such as influenza.

#64 · Patient Care

What PPE is used for airborne precautions and for which organisms?

Answer

A fitted N95 or higher respirator, ideally in a negative pressure room, used for organisms like tuberculosis, measles, and chickenpox.

#65 · Patient Care

What is neutropenic precautions?

Answer

Also called reverse isolation, it protects an immunocompromised patient with very low white blood cell counts from the staff and environment, essentially flipping standard precautions to protect the patient rather than the worker.

#66 · Patient Care

What is a nosocomial infection?

Answer

Also called a healthcare-associated infection, it is an infection a patient acquires during their stay in a healthcare facility that they did not have on admission.

#67 · Patient Care

What is a Safety Data Sheet (SDS)?

Answer

A document, formerly called an MSDS, that lists a substance's hazards, safe handling and storage instructions, required PPE, and first aid measures in case of exposure; one must be available for every hazardous chemical in the department.

#68 · Patient Care

How should hazardous chemicals be disposed of in a radiology department?

Answer

Through the correct waste stream, never poured down a standard sink drain unless specifically approved.

#69 · Patient Care

What does premedication for a contrast study usually involve?

Answer

A steroid and antihistamine regimen ordered in advance for a patient with a known contrast allergy history.

#70 · Patient Care

Why does a shellfish allergy matter for iodinated contrast screening?

Answer

It does not indicate true iodine sensitivity, since iodine itself is not an allergen and shellfish allergens do not cross-react with iodinated contrast, but it is still screened for because any history of significant allergic reactions, to shellfish or anything else, is linked to a modestly higher general risk of a contrast reaction.

#71 · Patient Care

Why should a non-contrast exam be performed before a barium exam when multiple studies are scheduled?

Answer

Retained barium can obscure other studies for days.

#72 · Patient Care

What is the difference between ionic and non-ionic iodinated contrast?

Answer

Ionic contrast dissociates into charged particles and has higher osmolality, associated with a higher rate of adverse reactions, while non-ionic contrast does not dissociate, has lower osmolality, and is generally safer despite higher cost.

#73 · Patient Care

When is barium sulfate contraindicated?

Answer

When bowel perforation is suspected, because free barium in the peritoneal cavity can cause a severe inflammatory reaction called barium peritonitis; a water-soluble iodinated agent is used instead.

#74 · Patient Care

What lab value is most directly used to screen for contrast-induced nephropathy risk?

Answer

eGFR, estimated glomerular filtration rate.

#75 · Patient Care

What is extravasation?

Answer

Contrast leaking out of the vein into surrounding tissue, causing swelling and pain.

#76 · Patient Care

What is phlebitis?

Answer

Inflammation of the vein itself, one of the local complications of contrast injection.

#77 · Patient Care

What are examples of a mild contrast reaction and how is it managed?

Answer

Nausea, a few hives, or a feeling of warmth, usually requiring only observation and reassurance.

#78 · Patient Care

What are examples of a moderate contrast reaction and how is it managed?

Answer

More extensive hives, wheezing, or vomiting, requiring close monitoring and possibly antihistamine treatment.

#79 · Patient Care

What are examples of a severe contrast reaction and how is it managed?

Answer

Laryngeal edema, severe bronchospasm, loss of consciousness, or cardiac arrest, requiring immediate emergency response including epinephrine, oxygen, and activation of the emergency response system.

#80 · Patient Care

What should a radiographer do first if a contrast reaction is recognized while the injection is still running?

Answer

Stop the injection, call for help immediately, and stay with the patient.

#81 · Safety

What is thermionic emission?

Answer

The process by which the heated tungsten cathode filament releases free electrons when current raises its temperature to roughly 2,200 degrees Celsius.

#82 · Safety

What is the space charge?

Answer

The cloud of free electrons that collects near the heated cathode filament before the exposure switch is pressed.

#83 · Safety

What does the focusing cup do?

Answer

It is a negatively charged structure surrounding the filament that repels the electron cloud and shapes it into a narrow stream aimed at the focal spot on the anode.

#84 · Safety

What percentage of electron kinetic energy converts to x-ray photons at the anode?

Answer

Fewer than 1 percent converts into x-ray photons, while more than 99 percent converts to heat, which is why rotating anodes and cooling systems are essential.

#85 · Safety

What is bremsstrahlung radiation?

Answer

X-ray photons produced when an incoming electron is deflected and slowed by the nucleus of a tungsten atom, with the lost energy emitted as a photon.

#86 · Safety

Why does bremsstrahlung radiation produce a continuous energy spectrum?

Answer

Because the amount of energy an electron loses depends on how close it passes to the nucleus and how much its path bends, so photon energies vary continuously rather than taking one fixed value.

#87 · Safety

What is characteristic radiation?

Answer

A photon released with a fixed, discrete energy when an incoming electron ejects an inner shell electron from a tungsten atom and an outer shell electron drops down to fill the vacancy.

#88 · Safety

At what kVp does characteristic radiation begin to appear in a tungsten target beam?

Answer

The kVp must exceed the K-shell binding energy of tungsten, approximately 69.5 keV, so characteristic x-rays appear only once kVp is set above roughly 70.

#89 · Safety

How are x-ray wavelength, frequency, and photon energy related?

Answer

Wavelength and frequency are inversely related, and since energy is directly proportional to frequency, a shorter wavelength always means a higher photon energy.

#90 · Safety

What controls x-ray beam quality?

Answer

Beam quality, the penetrating power of the beam, is controlled primarily by kVp and is measured using half-value layer.

#91 · Safety

What is half-value layer (HVL)?

Answer

The thickness of a specified material needed to reduce the intensity of the x-ray beam by half, used as the measure of beam quality.

#92 · Safety

What controls x-ray beam quantity?

Answer

Beam quantity, the total number of photons in the beam, is controlled primarily by mAs, though raising kVp also increases quantity as a side effect.

#93 · Safety

What is primary radiation?

Answer

The useful beam as it leaves the x-ray tube and travels toward the patient, carrying the energy that will form the image.

#94 · Safety

What is remnant (exit) radiation?

Answer

The portion of the primary beam remaining after passing through the patient, having lost photons to absorption and scatter, and it is this beam that reaches the image receptor.

#95 · Safety

State the inverse square law formula.

Answer

$I_1 \text{ times } D_1^2 \text{ equals } I_2 \text{ times } D_2^2$, where I is intensity and D is distance from the source.

#96 · Safety

Calculation: Exposure is 100 mR at 40 inches SID. What is the exposure at 80 inches SID?

Answer

Doubling the distance from 40 to 80 inches cuts intensity to one quarter, so the exposure becomes 25 mR.

#97 · Safety

Calculation: Exposure is 20 mR at 72 inches. What is the exposure at 36 inches?

Answer

Halving the distance from 72 to 36 inches quadruples the intensity, so the exposure becomes 80 mR.

#98 · Safety

Calculation: A beam measures 400 mR at 36 inches. What is the intensity at 72 inches?

Answer

Doubling the distance from 36 to 72 inches cuts intensity to one quarter, so the intensity becomes 100 mR.

#99 · Safety

What is the photoelectric effect?

Answer

A photon is completely absorbed by an inner shell electron, ejecting that electron from the atom and depositing all the photon's energy with no photon left to scatter.

#100 · Safety

How does photoelectric absorption relate to atomic number?

Answer

It increases dramatically with atomic number, varying with the cube of atomic number, which is why bone absorbs far more radiation than soft tissue and appears white on the image.

#101 · Safety

What is the Compton effect?

Answer

A photon strikes a loosely bound outer shell electron and ejects it, but instead of being absorbed the photon scatters off in a new direction carrying reduced energy.

#102 · Safety

What does Compton scattering depend on, and why does it matter for staff radiation safety?

Answer

It depends mainly on tissue electron density rather than atomic number, and since it dominates in soft tissue at diagnostic energies, it is the main source of occupational scatter exposure to radiographers.

#103 · Safety

What is coherent (Thomson) scattering?

Answer

A low energy photon causes an atom to vibrate briefly and is re-emitted in a new direction at essentially the same energy, with no ionization occurring.

#104 · Safety

What is attenuation?

Answer

The overall reduction in x-ray beam intensity as it passes through the patient, increasing with tissue thickness, density, and atomic number.

#105 · Safety

Which SI unit measures absorbed dose, and what is its older conventional equivalent?

Answer

Absorbed dose is measured in gray (Gy), expressed as joules per kilogram, and the older conventional unit is the rad, where 1 gray equals 100 rad.

#106 · Safety

Which SI unit measures dose equivalent, and what is its older conventional equivalent?

Answer

Dose equivalent is measured in sievert (Sv), and the older conventional unit is the rem, where 1 sievert equals 100 rem.

#107 · Safety

For diagnostic x-rays, how do gray and sievert compare numerically?

Answer

The radiation weighting factor for diagnostic x-rays is 1, so absorbed dose in gray and dose equivalent in sievert end up numerically equal.

#108 · Safety

Which unit measures exposure, and what does it describe?

Answer

Exposure is measured in coulombs per kilogram (C/kg) and describes the amount of ionization produced in a specific mass of air by the beam, with the roentgen as its older conventional unit.

#109 · Safety

What is effective dose, and what unit is it measured in?

Answer

Effective dose, measured in sievert, applies tissue weighting factors for each organ's radiosensitivity to produce one number representing the equivalent whole body risk from an uneven dose distribution.

#110 · Safety

What is air kerma?

Answer

A quantity measured in gray representing the kinetic energy released in a small mass of air at a given point, now the standard quantity fluoroscopic equipment reports for tracking patient skin dose.

#111 • Safety

Calculation: A patient receives an absorbed dose of 0.05 gray. What is that dose in rad?

Answer

Since 1 gray equals 100 rad, 0.05 gray equals 5 rad.

#112 • Safety

Calculation: A worker's dose equivalent for the year is 0.1 sievert. What is that in rem?

Answer

Since 1 sievert equals 100 rem, 0.1 sievert equals 10 rem.

#113 · Safety

What is linear energy transfer (LET)?

Answer

A measure of how much energy a type of radiation deposits per unit length as it travels through tissue, with alpha particles and neutrons producing high LET and diagnostic x-rays producing low LET.

#114 · Safety

What is relative biological effectiveness (RBE)?

Answer

A comparison of how much biological damage a given radiation type causes against a reference beam of 250 kVp x-rays delivering the same absorbed dose, with high LET radiation generally producing a higher RBE.

#115 · Safety

State the law of Bergonie and Tribondeau.

Answer

Cells are most radiosensitive when they divide frequently, are undifferentiated or immature, and have a long dividing future ahead of them.

#116 · Safety

Name four highly radiosensitive cell types according to the law of Bergonie and Tribondeau.

Answer

Blood-forming cells of the bone marrow, the lining of the gastrointestinal tract, reproductive cells, and the basal layer of the skin.

#117 · Safety

Which cell types are among the most radioresistant, and why?

Answer

Mature nerve cells and muscle cells, because they rarely if ever divide.

#118 · Safety

What is LD50/30, and what is its approximate value in humans?

Answer

The lethal dose to 50 percent of an exposed population within 30 days, approximately 3 to 4 gray delivered as a single whole body dose without medical treatment.

#119 · Safety

What is the oxygen effect in radiobiology?

Answer

Well-oxygenated cells are measurably more sensitive to radiation damage than hypoxic, poorly oxygenated cells, a principle relevant to radiation therapy planning for tumors with poor blood supply.

#120 · Safety

What defines a deterministic (tissue reaction) effect?

Answer

It has a threshold dose below which the effect does not occur, and once that threshold is exceeded, severity increases as dose increases. Skin erythema, cataracts, epilation, and sterility are examples.

#121 · Safety

What defines a stochastic effect?

Answer

There is no dose threshold, so any dose carries some probability of the effect occurring. As dose increases the probability rises, but the severity of the effect if it occurs does not change. Cancer and genetic effects are the two classic examples.

#122 · Safety

How do early and late radiation effects differ in timing?

Answer

Early effects appear within minutes to weeks of exposure, such as skin erythema or acute radiation syndrome, while late effects may not appear for years or decades, such as cataracts or cancer.

#123 · Safety

How does chronic exposure differ from acute exposure in biological effect?

Answer

The same total dose delivered chronically over a long period is generally far less harmful than the identical total dose delivered acutely, because cells have time to repair sublethal damage between fractions.

#124 · Safety

At what approximate dose range can the lens of the eye develop cataracts?

Answer

Roughly 0.5 to 2 gray, often with a latent period of years before the cataract becomes clinically apparent.

#125 • Safety

At what approximate doses do skin erythema and epilation occur?

Answer

Erythema appears around 2 gray, and epilation, meaning hair loss, occurs at higher doses in the range of 3 to 5 gray.

#126 • Safety

At what dose can temporary male sterility occur, and what dose causes permanent sterility?

Answer

Temporary sterility in males is possible from doses as low as approximately 0.15 gray, while permanent sterility in either sex requires several gray.

#127 · Safety

List the four stages of acute radiation syndrome.

Answer

A prodromal stage of nausea and malaise, a latent stage of temporary improvement, a manifest illness stage where organ damage becomes apparent, and finally recovery or death.

#128 · Safety

What is hematopoietic syndrome, and at what dose does it begin?

Answer

A pattern of illness starting around 1 gray whole body dose, caused by damage to bone marrow stem cells, leading to infection risk from low white cell counts and bleeding risk from low platelets.

#129 · Safety

What is gastrointestinal syndrome, and at what dose range does it occur?

Answer

A pattern of illness occurring at whole body doses of roughly 6 to 10 gray, caused by destruction of the intestinal lining, producing severe fluid loss and infection with death typically within one to two weeks.

#130 · Safety

What is central nervous system (cerebrovascular) syndrome, and at what dose does it occur?

Answer

A pattern of illness occurring at whole body doses generally above 50 gray, causing cerebral edema and vascular collapse, with death typically within hours to a few days. It is the fastest and most severe acute radiation syndrome.

#131 · Safety

What happens if radiation exposure occurs during the preimplantation stage of pregnancy, roughly the first two weeks after conception?

Answer

The prevailing model is all-or-none, meaning a significant dose either kills the embryo outright or the surviving cells compensate completely with no lasting defect.

#132 · Safety

Why is the organogenesis stage of pregnancy, roughly weeks two through eight, the period of greatest risk for congenital malformations?

Answer

The developing organs are actively forming during this stage, making radiation exposure most likely to cause structural birth defects.

#133 · Safety

During which gestational age window is the risk of radiation-induced mental retardation and microcephaly highest?

Answer

Roughly week eight to week fifteen, within the fetal stage, when the central nervous system is most vulnerable to growth and developmental effects.

#134 · Safety

How does increasing kVp affect patient skin dose?

Answer

Increasing kVp increases beam penetration, so fewer photons are needed to reach the receptor. This generally lowers patient skin dose, though it also lowers subject contrast.

#135 · Safety

How does increasing mAs affect patient dose?

Answer

Increasing mAs increases the number of photons in the beam and raises patient dose in a directly proportional way, without changing beam quality.

#136 · Safety

What does automatic exposure control (AEC) do?

Answer

AEC uses ionization chambers positioned between the patient and the image receptor to terminate the exposure once enough radiation has reached the detector, protecting against both overexposure and underexposure.

#137 · Safety

What is a common repeat cause related to AEC?

Answer

Selecting the wrong AEC chamber or mispositioning the patient over the chambers, which either cuts the exposure short or lets it run too long.

#138 · Safety

Why is tight collimation one of the most effective dose reduction tools?

Answer

Collimating to the smallest field size that covers the anatomy of interest reduces the volume of tissue irradiated and reduces scatter production, which lowers dose and improves image contrast.

#139 · Safety

Why is a repeat exposure especially costly in terms of dose?

Answer

A repeat is effectively a doubling of dose for that view, since the patient receives the full exposure twice.

#140 · Safety

Why do pediatric patients need reduced technique?

Answer

Their tissues are more radiosensitive and a smaller body part requires less penetration to produce a diagnostic image.

#141 • Safety

Why are immobilization devices preferred over manually holding a pediatric patient?

Answer

Immobilization devices eliminate motion without requiring anyone to be in the room, avoiding unnecessary occupational exposure to a family member or staff.

#142 • Safety

How does larger body habitus affect exposure technique?

Answer

A larger body habitus generally requires higher kVp and mAs to penetrate adequately, which increases patient dose compared to an average-sized patient.

#143 · Safety

What special considerations apply to imaging a pregnant patient?

Answer

Shielding when appropriate, tight collimation, and confirming clinical necessity before proceeding, since the fetus is especially sensitive to radiation during organogenesis.

#144 · Safety

What is the purpose of filtration in the x-ray beam?

Answer

Filtration removes the lowest energy photons before they reach the patient. Those photons would be absorbed in skin and superficial tissue without contributing to the image, so removing them reduces skin and organ dose.

#145 · Safety

What is beam hardening?

Answer

The rise in average beam energy that results from filtration removing the lowest energy photons, leaving a beam with fewer low-energy photons and a higher average energy.

#146 · Safety

What is the NCRP Report 102 minimum total filtration for a general purpose unit above 70 kVp?

Answer

2.5 millimeters aluminum equivalent, with lower minimums permitted at lower operating kVp ranges.

#147 · Safety

What makes up total beam filtration?

Answer

A combination of inherent filtration from the tube housing and window, plus added filtration, usually aluminum sheets placed in the collimator housing.

#148 · Safety

Why does modern digital equipment record an exposure index for each image?

Answer

It supports quality assurance and regulatory compliance by documenting the actual radiation exposure delivered for each digital image, and many facilities log this patient dose data as part of the permanent record.

#149 · Safety

What is detective quantum efficiency (DQE)?

Answer

A measure of how efficiently a digital detector converts incoming x-ray photons into usable image signal. A higher DQE detector can produce a diagnostic image with fewer photons, meaning lower patient dose.

#150 · Safety

Why is overexposure a hidden risk with digital imaging?

Answer

Digital receptors have wide exposure latitude, so an image can look acceptable across a broad range of exposures, meaning it can look fine even when the patient received far more dose than necessary. The exposure indicator must be actively monitored.

#151 · Safety

Why does using a grid increase patient dose?

Answer

A higher grid ratio requires higher mAs to compensate for radiation the grid absorbs, including some useful primary beam along with the scatter.

#152 · Safety

When is using a grid not justified?

Answer

On a body part too thin to generate clinically significant scatter, using a grid unnecessarily raises patient dose without a meaningful gain in image quality.

#153 · Safety

How does pulsed fluoroscopy reduce dose compared to continuous fluoroscopy?

Answer

Pulsed fluoroscopy delivers the beam in short pulses instead of a continuous stream, which can substantially lower dose at the same perceived frame rate since the eye perceives pulsed images as smooth motion.

#154 · Safety

Why does minimizing total fluoroscopy time matter so much for dose?

Answer

Dose accumulates continuously while the fluoroscopic beam is on, so reducing beam-on time is one of the most direct controls available for both patient and staff dose.

#155 · Safety

What does the fluoroscopic cumulative timer do?

Answer

It sounds an audible signal at preset intervals, typically every five minutes of beam-on time, to remind the operator how long the beam has actually been active.

#156 · Safety

What does automatic brightness control (ABC) do in fluoroscopy?

Answer

It automatically adjusts kVp and mA to maintain consistent image brightness as tissue thickness or density changes under the beam, similar in concept to AEC in radiography.

#157 · Safety

How does image receptor positioning affect fluoroscopic dose?

Answer

Keeping the receptor or image intensifier as close to the patient as possible reduces the tissue the beam must traverse and improves image quality, allowing a lower technique.

#158 · Safety

Why does fluoroscopic magnification mode increase patient dose?

Answer

Magnification mode increases spatial resolution but requires a higher mA to maintain adequate photon flux at the smaller field of view, so it should only be used when diagnostic benefit justifies the dose cost.

#159 · Safety

What is last image hold and why does it reduce dose?

Answer

It displays the final fluoroscopic image on the monitor after the beam turns off, letting the physician review it without keeping the beam active, avoiding re-activation of the beam just to look again.

#160 · Safety

What is the minimum source-to-skin distance for a stationary fluoroscopic unit?

Answer

38 centimeters, as regulated under 21 CFR, the federal performance standard for fluoroscopic devices.

#161 · Safety

What is the minimum source-to-skin distance for a mobile fluoroscopic unit such as a C-arm?

Answer

30 centimeters, as regulated under 21 CFR.

#162 · Safety

What is a dose area product (DAP) meter?

Answer

A device that measures the product of dose and beam area, giving a single value that reflects total energy delivered to the patient across an entire fluoroscopic procedure, useful for comparing and tracking cumulative dose.

#163 · Safety

What are the Image Wisely and Image Gently campaigns?

Answer

Image Wisely focuses on reducing unnecessary radiation exposure in adult imaging, and Image Gently focuses specifically on pediatric imaging. Both provide practical guidance and protocols supporting dose reduction.

#164 · Safety

What does ALARA stand for and what does it apply to?

Answer

As Low As Reasonably Achievable. Per ARRT's guidance, ALARA applies specifically to personnel protection, not patient protection.

#165 · Safety

How is patient dose managed differently from personnel dose under this framework?

Answer

Patient dose is managed through exposure factor selection, filtration, and collimation. ALARA is the guiding philosophy for keeping occupational dose to staff as low as reasonably achievable.

#166 · Safety

What is a common student mistake regarding ALARA?

Answer

Blending ALARA together with patient protection principles, when ALARA is actually specific to occupational (personnel) protection.

#167 · Safety

What are the three sources of radiation exposure to personnel in an x-ray suite?

Answer

The primary x-ray beam, secondary radiation (scatter and leakage), and specifically the patient as the largest source of scatter in the room.

#168 · Safety

Why is the primary x-ray beam considered the most dangerous source of personnel exposure?

Answer

It carries the highest intensity radiation in the room, and personnel should never have any part of their body in the primary beam.

#169 · Safety

What is scatter radiation?

Answer

Radiation that has changed direction and lost some energy after interacting with matter, such as the patient's tissue.

#170 · Safety

What is leakage radiation?

Answer

Radiation that escapes through the x-ray tube housing itself despite the housing's shielding.

#171 · Safety

Why is the patient the largest source of scatter radiation in the room during an exposure?

Answer

The patient's own tissue is what most of the primary beam interacts with, so standing close to the patient during an exposure increases occupational dose even though the technologist is never in the primary beam.

#172 · Safety

What is the time principle of radiation protection?

Answer

The less time spent near a radiation source, the lower the cumulative dose received, since dose accumulates linearly with exposure duration.

#173 · Safety

What does the inverse square law state?

Answer

Radiation intensity is inversely proportional to the square of the distance from the source.

#174 · Safety

A technologist at 1 meter from a fluoroscopic tube receives 10 milligray per hour. What is the exposure rate at 2 meters?

Answer

2.5 milligray per hour, since 10 divided by 2 squared (4) equals 2.5, per the inverse square law.

#175 · Safety

A technologist at 1 meter from a fluoroscopic tube receives 10 milligray per hour. What is the exposure rate at 3 meters?

Answer

About 1.1 milligray per hour, since 10 divided by 3 squared (9) is approximately 1.1, per the inverse square law.

#176 · Safety

What is the shielding principle of radiation protection?

Answer

Placing attenuating material between a person and the radiation source, using devices like lead aprons, thyroid shields, lead glasses, and fixed or mobile barriers.

#177 · Safety

What is the minimum lead equivalent for a standard protective apron, and how much scatter does it attenuate?

Answer

0.5 millimeter lead equivalent per NCRP Report 102, attenuating roughly 90 to 95 percent of scatter radiation at typical diagnostic energies. Aprons must be inspected periodically for cracks that would defeat that attenuation.

#178 · Safety

What are the protection rules for mobile radiography exposures?

Answer

Retreat as far as the exposure cord and room allow, ideally a minimum of 2 meters from the patient and tube, and never hold a patient or image receptor during exposure except in genuine emergencies, wearing an apron and standing as far from the beam as possible if unavoidable.

#179 · Safety

What are the minimum lead equivalents for the fluoroscopic protective curtain and the Bucky slot cover?

Answer

Both are required to be at least 0.25 millimeter lead equivalent. The curtain hangs between the fluoroscopist and patient, and the Bucky slot cover automatically shields the Bucky slot opening when the spot film device is withdrawn.

#180 · Safety

What are the requirements for the fluoroscopic high level control and exposure switch?

Answer

The high level control switch requires continuous manual engagement and cannot latch on unintentionally. The exposure switch itself must be a dead man type switch, so exposure continues only while it is actively held and stops the instant it is released.

#181 • Safety

What is the advantage of remote controlled fluoroscopy?

Answer

It allows the operator to step out of the room entirely for certain procedures, the most effective distance based protection available for that equipment type.

#182 • Safety

What are the three main types of personnel dosimeters?

Answer

The thermoluminescent dosimeter (TLD), the optically stimulated luminescence dosimeter (OSL), and the direct ion storage (DIS) dosimeter.

#183 · Safety

How does a TLD dosimeter work?

Answer

It releases trapped energy as light when heated after exposure, and the amount of light released is proportional to the dose received.

#184 · Safety

How does an OSL dosimeter work?

Answer

It releases trapped energy as light when stimulated with a specific laser wavelength, and unlike a TLD it can be reanalyzed.

#185 · Safety

How does a DIS dosimeter work?

Answer

It uses an ion chamber design that allows a real time readout of dose, unlike TLD or OSL devices.

#186 · Safety

Where should a dosimeter be worn, and when is a second one used?

Answer

The primary dosimeter is worn consistently at collar level outside the lead apron, to reflect dose to unshielded areas like the head, neck, and lens of the eye. A second dosimeter is sometimes worn under the apron at waist level for pregnant workers or for precise organ dose estimation.

#187 · Safety

What are examples of improper dosimeter handling?

Answer

Wearing it upside down, sharing it between individuals, or leaving it somewhere it might record artificial exposure, such as near an x-ray unit or in a hot car.

#188 · Safety

What is the annual effective dose limit for occupationally exposed radiation workers?

Answer

50 millisievert, equivalent to 5 rem in the older conventional unit.

#189 · Safety

How is the lifetime cumulative effective dose limit for a radiation worker calculated?

Answer

The worker's age in years multiplied by 10 millisievert, equivalent to 1 rem per year of age.

#190 · Safety

What is the annual dose limit for members of the public?

Answer

1 millisievert, equivalent to 0.1 rem, reflecting that the public has not chosen occupational exposure and receives no direct benefit from it.

#191 · Safety

What is the monthly dose limit for a declared pregnant worker's embryo or fetus?

Answer

0.5 millisievert per month, equivalent to 0.05 rem per month, once the worker has formally declared her pregnancy in writing.

#192 · Safety

What is the total gestational dose limit goal for a declared pregnant worker?

Answer

Keeping total fetal dose for the entire pregnancy under 5 millisievert, equivalent to 0.5 rem.

#193 · Safety

How must facilities handle dosimetry records and readings that approach an investigational level?

Answer

Dosimetry records must be evaluated regularly and maintained as permanent occupational records documenting compliance and career dose, and any reading approaching or exceeding an investigational level must be promptly reviewed, investigated, and documented.

#194 · Safety

How should short lived radioactive material generally be handled before disposal?

Answer

It should decay in secured storage behind adequate shielding before being disposed of as ordinary waste.

#195 · Safety

What must be confirmed before radioactive material or an area is released from radioactive material control?

Answer

Documented survey and wipe testing to confirm no contamination remains.

#196 · Image Production

What is receptor exposure and what factor controls it primarily?

Answer

Receptor exposure is the amount of radiation reaching the image receptor. It is controlled primarily by mAs, which has a direct, straight-line relationship with the number of photons produced at the tube.

#197 • Image Production

What does mAs control, and what does it NOT affect?

Answer

mAs controls receptor exposure only. It has no effect on contrast, spatial resolution, or distortion because it does not change beam energy or geometry.

#198 • Image Production

How is mAs calculated?

Answer

mAs equals mA (tube current) multiplied by exposure time in seconds. For example, 200 mA at 0.05 seconds equals 10 mAs.

#199 • Image Production

What is the 15 percent rule?

Answer

Increasing kVp by 15 percent roughly doubles receptor exposure. The standard dose-reduction technique is to raise kVp 15 percent and cut mAs in half, keeping receptor exposure about the same while lowering patient dose.

#200 • Image Production

List all the image quality categories that kVp affects.

Answer

kVp raises receptor exposure, lowers image contrast (longer gray scale), and lowers subject contrast. It does not directly affect spatial resolution or distortion.

#201 • Image Production

Why does raising kVp lower image contrast?

Answer

Higher kVp produces more scatter relative to primary photons, which fogs the image and produces a longer gray scale with more shades of gray.

#202 • Image Production

Why does raising kVp lower subject contrast?

Answer

A higher-energy beam is less selectively absorbed by dense structures like bone versus soft tissue, so the true difference in absorption between tissues shrinks.

#203 • Image Production

Calculation: mA is 300 and time is 0.04 seconds. What is the mAs?

Answer

300 times 0.04 equals 12 mAs.

#204 • Image Production

Calculation: a technique of 200 mA at 0.05 sec gives 10 mAs. Time is cut to 0.025 sec to reduce motion, halving receptor exposure. What mA restores the original 10 mAs?

Answer

mA must be doubled to 400, since 400 times 0.025 equals 10 mAs, restoring the original receptor exposure at the shorter exposure time.

#205 • Image Production

What does focal spot size control?

Answer

Focal spot size affects spatial resolution only. A smaller focal spot produces less penumbra and a sharper image, but has no effect on receptor exposure, contrast, or distortion.

#206 • Image Production

Why is a larger focal spot sometimes used for thick body parts despite lower resolution?

Answer

A larger focal spot tolerates higher heat loading, which is needed for the higher mA and longer exposure times that thick parts often require, trading some sharpness for tube protection.

#207 • Image Production

What image quality factor does motion affect?

Answer

Motion affects spatial resolution only. It blurs recorded detail but does not change receptor exposure, contrast, or distortion.

#208 • Image Production

Describe the anode heel effect and what it affects.

Answer

The anode heel effect affects receptor exposure only. Because the anode is angled, beam intensity is lower toward the anode end of the tube and higher toward the cathode end.

#209 • Image Production

How do you use the anode heel effect for positioning a chest or hip image?

Answer

Place the thicker part of the anatomy, such as the thoracic spine or the femur, toward the cathode end of the tube, where exposure is higher, to help even out density across the image.

#210 • Image Production

What three image quality categories does SID affect?

Answer

SID affects receptor exposure, spatial resolution, and distortion.

#211 • Image Production

What law governs how SID affects receptor exposure?

Answer

The inverse square law: intensity is inversely proportional to the square of the distance from the source.

#212 • Image Production

Calculation: a technique of 10 mAs is adequate at 40 inches SID. What mAs is needed at 72 inches?

Answer

New mAs equals old mAs times (new distance squared divided by old distance squared): 10 times (72 squared divided by 40 squared) equals 10 times 3.24, or 32.4 mAs.

#213 • Image Production

How does SID affect spatial resolution and distortion?

Answer

A longer SID reduces penumbra, sharpening the image, and produces less magnification of the part, reducing size distortion.

#214 • Image Production

What three image quality categories does OID affect, and how does that differ from SID?

Answer

OID affects image contrast, spatial resolution, and distortion, but unlike SID it does not directly control receptor exposure.

#215 • Image Production

Why does increasing OID typically improve image contrast?

Answer

Moving the part farther from the receptor lets scattered photons diverge away from the receptor and miss it, the air-gap effect, which typically improves rather than degrades contrast.

#216 • Image Production

What is the general positioning rule for OID?

Answer

Keep OID as small as possible for every projection unless magnification is being deliberately used, such as in some cross-table lateral projections.

#217 • Image Production

Define size distortion and shape distortion.

Answer

Size distortion is magnification. Shape distortion is foreshortening or elongation caused by misalignment of the central ray, tube, or part.

#218 • Image Production

How is magnification calculated?

Answer

Magnification equals SID divided by SOD (source-to-object distance), where SOD equals SID minus OID.

#219 • Image Production

Differentiate foreshortening from elongation.

Answer

Foreshortening occurs when a structure is angled away from the receptor plane so it appears shorter than it actually is. Elongation occurs when the angulation instead stretches the structure so it appears longer than its true length.

#220 • Image Production

What does tube, part, or receptor angulation affect?

Answer

Angulation affects receptor exposure, spatial resolution, and distortion, since misalignment changes beam path length, changes the geometry controlling sharpness, and directly creates shape distortion.

#221 • Image Production

What two image quality categories do grids affect?

Answer

Grids affect receptor exposure and image contrast.

#222 • Image Production

How does a grid improve image contrast?

Answer

A grid is made of thin lead strips that absorb scatter radiation before it reaches the receptor, cleaning up the image and improving contrast.

#223 • Image Production

Why does adding a grid require a technique increase?

Answer

The grid absorbs some of the primary useful beam along with the scatter, lowering receptor exposure, so mAs must be increased to compensate.

#224 • Image Production

What is the grid conversion factor also called, and what does it tell you?

Answer

It is also called the Bucky factor. It tells you how many times to multiply the non-grid mAs to get an equivalent grid technique.

#225 • Image Production

Calculation: a non-grid technique of 4 mAs is adequate, and a 12:1 grid with a conversion factor of 5x is added. What is the new mAs?

Answer

4 times 5 equals 20 mAs.

#226 • Image Production

What is the approximate grid conversion factor for an 8:1 grid?

Answer

About 4 times the non-grid mAs.

#227 • Image Production

What three image quality categories does tube filtration affect?

Answer

Filtration affects receptor exposure, image contrast, and subject contrast.

#228 • Image Production

What does added filtration do to the x-ray beam?

Answer

Added aluminum filtration absorbs the lowest-energy photons that would otherwise be absorbed by the patient's skin without reaching the image, hardening the beam and raising the average photon energy.

#229 • Image Production

What is the minimum total filtration requirement above 70 kVp?

Answer

Total filtration, inherent plus added, must generally equal at least 2.5 millimeters aluminum equivalent for a fixed x-ray unit operating above 70 kVp.

#230 • Image Production

What two image quality categories does beam restriction (collimation) affect?

Answer

Beam restriction affects receptor exposure and image contrast.

#231 • Image Production

Why does tighter collimation improve contrast?

Answer

Tightening the field reduces the tissue volume irradiated, which reduces scatter produced within the patient. Less scatter reaching the receptor means better image contrast.

#232 • Image Production

What four image quality categories do patient factors like size, density, and pathology affect?

Answer

Patient factors affect receptor exposure, image contrast, subject contrast, and spatial resolution, since a larger or denser patient attenuates more beam and produces more internal scatter.

#233 • Image Production

Give examples of additive pathology and how they change technique.

Answer

Ascites, pneumonia, edema, and a plaster cast are additive conditions. They add tissue density and require a technique increase.

#234 • Image Production

Give examples of destructive pathology and how they change technique.

Answer

Emphysema, osteoporosis, and a resolving fracture are destructive conditions. They remove tissue and require a technique decrease.

#235 • Image Production

What is a fixed kVp technique chart system?

Answer

A system where kVp is set at an optimal level for each body part and only mAs is adjusted as thickness changes, keeping contrast consistent since kVp never changes.

#236 • Image Production

What is a variable kVp technique chart system?

Answer

A system where both kVp and mAs move together as thickness changes, allowing lower dose on thinner parts but producing less consistent contrast between exams.

#237 • Image Production

What is anatomically programmed technique?

Answer

A generator feature that lets the technologist select the body part and projection from a menu. The generator loads a preset kVp, mA, and time based on average patient measurements, which the technologist fine-tunes using a caliper measurement of part thickness.

#238 • Image Production

How much should mAs typically be increased for a plaster cast versus a fiberglass cast?

Answer

A plaster cast is highly additive and typically needs mAs increased by roughly 50 to 60 percent. A fiberglass cast needs little adjustment since it attenuates far less.

#239 • Image Production

How should technique generally be adjusted for pediatric patients?

Answer

Pediatric patients generally need lower mAs and often higher kVp with very short exposure times to limit motion, since children move unpredictably and their tissues are thinner.

#240 • Image Production

How should technique generally be adjusted for geriatric patients?

Answer

Geriatric patients often have reduced tissue mass and bone density from osteoporosis, so they usually need reduced technique along with extra time to prevent motion given limited mobility.

#241 • Image Production

How does contrast media affect exposure technique?

Answer

Contrast media such as barium or iodinated contrast increases attenuating density and typically needs a technique increase similar to an additive pathology.

#242 • Image Production

What does AEC (automatic exposure control) actually control?

Answer

AEC terminates the exposure automatically once a preset amount of radiation has reached the detector cells, rather than the technologist manually setting exposure time.

#243 • Image Production

Under AEC, what happens to receptor exposure when the mA station is changed?

Answer

Receptor exposure stays constant regardless of the mA station selected, as long as mA is high enough that exposure time does not fall below the system's minimum response time.

#244 • Image Production

Under AEC, how does raising kVp behave differently than under manual technique?

Answer

Raising kVp still shortens contrast and lengthens gray scale as it does manually, but AEC cuts the exposure short once enough radiation is detected, so higher kVp typically shortens exposure time and can reduce dose while barely changing receptor exposure.

#245 • Image Production

What is the classic boards scenario for AEC detector cell misuse on a chest projection?

Answer

If the active cells sit under the lungs rather than the mediastinum, the AEC senses the low-density lung field and terminates the exposure too soon, causing gross underexposure of the heart and great vessels.

#246 • Image Production

What is the exposure adjustment (density) control on AEC used for?

Answer

It lets the technologist intentionally raise or lower the receptor exposure the AEC is targeting, typically in increments like plus one or minus one, each representing roughly 25 to 50 percent change, useful when patient tissue characteristics would otherwise fool the AEC.

#247 • Image Production

What is AEC backup time or backup mAs?

Answer

A maximum exposure the unit allows regardless of what the detector cells detect, protecting the patient and tube if a positioning error or malfunction prevents proper termination.

#248 • Image Production

What controls spatial resolution in a digital detector?

Answer

Pixel size and pixel pitch, along with the size and fill factor of the underlying detector elements (DELs). Smaller DELs with tighter pitch and higher fill factor produce higher spatial resolution.

#249 · Image Production

What is the Nyquist frequency?

Answer

The highest spatial frequency a system can accurately record without aliasing artifacts. It equals one half of the sampling frequency set by pixel pitch.

#250 · Image Production

For a fixed field of view, how does matrix size affect spatial resolution?

Answer

A larger matrix means smaller pixels, which produces better spatial resolution.

#251 • Image Production

What does modulation transfer function (MTF) measure?

Answer

MTF, expressed as a value between 0 and 1, measures how faithfully a system reproduces true object contrast as spatial frequency increases. It is the standard objective measure of detector sharpness.

#252 • Image Production

What controls contrast resolution in a digital image?

Answer

Bit depth, the number of gray shades each pixel can represent. A higher bit depth, such as 12-bit versus 8-bit, displays far more subtle tissue density differences.

#253 • Image Production

What is detective quantum efficiency (DQE)?

Answer

DQE measures how efficiently a detector converts incoming photons into a usable, low-noise signal. A higher DQE means diagnostic quality using fewer photons, lowering patient dose.

#254 • Image Production

Differentiate a focused grid from a virtual grid.

Answer

A focused grid has lead strips angled to match beam divergence at a specific SID. A virtual grid applies the grid effect through post-processing, but it cannot remove scatter that was never eliminated at the time of exposure.

#255 • Image Production

What is quantum noise, also called quantum mottle?

Answer

The random, grainy appearance caused by too few photons reaching the detector to form a statistically reliable image.

#256 • Image Production

Why is checking image brightness not a reliable way to catch underexposure on a digital image?

Answer

Processing software automatically applies contrast and brightness adjustments to whatever signal it receives, so a low-mAs digital image can display as bright and sharp on the monitor even though it is underexposed. The only reliable check is the exposure indicator number.

#257 • Image Production

What are the risks of an underexposed digital image that looks fine on screen?

Answer

It still carries excess quantum mottle and reduced contrast resolution, and if the exposure indicator is not checked, the patient may need an avoidable repeat later.

#258 • Image Production

What are the rules for placing radiographic identification markers?

Answer

Right, left, and any positional markers must be placed within the collimated field at the time of exposure and never added digitally afterward, since a marker added after exposure cannot be verified against the patient's actual position.

#259 • Image Production

List the criteria checked when evaluating a completed radiographic image.

Answer

Exposure indicator, contrast (subject and image), spatial resolution, distortion (size and shape), identification markers present and correct, and freedom from artifacts such as motion, grid lines, or processing errors.

#260 • Image Production

What does the x-ray generator do?

Answer

It takes incoming electrical power from the wall and converts it into the precise, controlled voltage and current the x-ray tube needs to produce an x-ray beam, using transformers and a rectification system.

#261 • Image Production

What is the job of transformers in the x-ray circuit?

Answer

Transformers change voltage up or down through electromagnetic induction, stepping voltage up for the tube and stepping filament voltage down for the cathode.

#262 • Image Production

What does rectification do in the x-ray circuit?

Answer

Rectification converts alternating current into direct current so electrons keep flowing in one direction, from cathode to anode, instead of reversing direction.

#263 • Image Production

How does a single-phase generator's waveform differ from a three-phase generator's waveform?

Answer

Single-phase generators produce a pulsating waveform with big dips between pulses, so the tube is not always at peak kilovoltage. Three-phase generators smooth that waveform out, keeping voltage much closer to a constant peak.

#264 • Image Production

How does a high-frequency generator achieve nearly constant voltage?

Answer

It converts incoming current to a very high frequency, sometimes over 100,000 Hz, before rectifying it, producing an almost constant voltage with minimal ripple.

#265 • Image Production

What practical advantages do high-frequency generators offer over single-phase units?

Answer

They are more efficient, produce more consistent image quality, and allow shorter exposure times for the same technique.

#266 • Image Production

What is tube loading?

Answer

Tube loading refers to how much heat a given exposure puts into the anode and how quickly that heat can be dissipated.

#267 • Image Production

Why do generators use tube rating charts?

Answer

To prevent the technologist from selecting a technique that would put more heat into the anode than the tube can handle, which would overload and damage the tube.

#268 • Image Production

What is the function of the cathode in an x-ray tube?

Answer

The cathode is the electron source. It contains a filament, usually tungsten, that is heated to release electrons through thermionic emission.

#269 • Image Production

What is the function of the anode in an x-ray tube?

Answer

The anode is the target that electrons strike to produce x-rays.

#270 • Image Production

Why is the anode target typically made of tungsten alloyed with rhenium?

Answer

Tungsten has a high atomic number, which is efficient for producing x-rays, and a high melting point, which lets it survive the intense heat generated during exposure.

#271 • Image Production

Why do rotating anode tubes spin the anode disc during exposure?

Answer

An induction motor spins the anode so the electron beam does not stay focused on one small spot. This spreads out the heat load and lets the tube tolerate higher techniques without melting the target.

#272 • Image Production

What is inherent filtration and what does it include?

Answer

Inherent filtration is filtration built into the tube itself, including the glass or metal tube housing and the surrounding oil.

#273 • Image Production

Why does filtration matter for patient dose?

Answer

Filtration removes the lowest-energy photons from the beam before they reach the patient. Those photons would be absorbed by the patient's skin and add dose without ever reaching the image receptor, so filtration lowers skin dose without meaningfully changing image quality.

#274 • Image Production

How does manual exposure control end an exposure?

Answer

The exposure ends when the preset time or mAs selected by the technologist is reached, regardless of the actual dose the patient received.

#275 • Image Production

What do the ionization chambers in an AEC system actually measure?

Answer

They measure, in real time, how much radiation has passed through the patient and reached the detector position between the patient and the image receptor.

#276 • Image Production

How does AEC terminate an exposure?

Answer

Once the ionization chambers accumulate a preset amount of radiation, they signal the generator to terminate the exposure.

#277 • Image Production

Why can AEC produce a properly exposed image on patients of very different body sizes using the same chamber selection?

Answer

Because AEC terminates based on radiation actually reaching the detector, it automatically compensates for differences in patient thickness and tissue density.

#278 • Image Production

What is the purpose of the backup timer built into every AEC circuit?

Answer

If the detector fails or the chambers never receive enough radiation to trigger termination, such as from positioning error placing tissue outside the selected chamber, the backup timer cuts off the exposure at a preset maximum time or mAs, protecting the patient from excessive dose and the tube from overload.

#279 • Image Production

What does the exposure adjustment control on the console do?

Answer

It lets the technologist bias the AEC to produce a lighter or darker image than the standard calibration, which is useful for patients with plaster casts, contrast media, or unusual pathology.

#280 • Image Production

What is the AEC minimum response time, and why does it matter for thin body parts?

Answer

Minimum response time is the shortest exposure time the AEC system is physically capable of producing. If a technique calls for a shorter exposure than that minimum, the AEC cannot terminate fast enough and the image will be overexposed, which is a real risk with thin body parts, pediatric patients, and high-mA techniques.

#281 • Image Production

How do direct conversion digital detectors work?

Answer

Direct conversion detectors, most commonly amorphous selenium, convert x-ray photons directly into an electrical signal without an intermediate light step.

#282 • Image Production

Why do direct conversion detectors have excellent inherent spatial resolution?

Answer

Because there is no light-conversion step, there is no lateral spread of light, which preserves sharp spatial detail.

#283 • Image Production

How do indirect conversion digital detectors work?

Answer

Indirect conversion detectors, using materials like amorphous silicon combined with a cesium iodide scintillator, first convert x-rays into visible light and then convert that light into an electrical signal.

#284 • Image Production

Why can indirect conversion detectors have slightly lower spatial resolution than direct conversion detectors?

Answer

The extra light-conversion step introduces some light spread, though modern cesium iodide's needle-like crystal structure channels light efficiently and minimizes this effect.

#285 • Image Production

Why does beam restriction with collimators reduce patient dose?

Answer

Restricting the field size to only the anatomy of interest reduces the volume of tissue irradiated, which reduces patient dose.

#286 • Image Production

Why does beam restriction with collimators improve image contrast?

Answer

A smaller field size reduces the amount of scatter radiation reaching the image receptor, and less scatter means better image contrast.

#287 • Image Production

How does an image intensifier produce a fluoroscopic image?

Answer

It converts x-rays into light at an input phosphor, converts that light into electrons, accelerates and focuses those electrons through a vacuum tube, and converts them back into a much brighter light image at the output phosphor.

#288 • Image Production

How do flat panel detectors used in fluoroscopy differ from image intensifiers?

Answer

Flat panel detectors skip the entire electron-optic chain and convert x-rays into a digital signal directly, similar to digital radiography detectors, which reduces equipment bulk and eliminates certain image intensifier distortion artifacts.

#289 • Image Production

What does automatic brightness control (ABC), also called automatic exposure rate control, do during fluoroscopy?

Answer

It operates continuously throughout the fluoroscopic exposure, automatically adjusting kVp, mA, or both, to maintain consistent image brightness on the monitor as the body part or positioning changes.

#290 • Image Production

What does magnification mode do in fluoroscopy?

Answer

It narrows the electron beam's focus within the image intensifier to display a smaller field of view at higher apparent resolution, useful for evaluating fine detail such as small vessels during an angiographic procedure.

#291 • Image Production

Why does magnification mode increase patient dose in fluoroscopy?

Answer

Because ABC must increase technique to maintain brightness over a smaller receptor area, even though the anatomical area actually being irradiated stays roughly the same.

#292 • Image Production

What is a grid and what does it do?

Answer

A grid is a device placed between the patient and the image receptor that absorbs scattered radiation before it reaches the receptor, which improves image contrast.

#293 • Image Production

Why do stationary grids typically use a higher grid frequency than moving grids?

Answer

Because a stationary grid's lead strips are visible as fine lines on the finished image, a higher grid frequency, meaning more lead strips per inch, keeps those lines from being visually distracting.

#294 • Image Production

What is a virtual grid?

Answer

A virtual grid is a software-based scatter correction used with digital detectors that mathematically reduces the effect of scatter on the displayed image instead of physically absorbing it with lead strips.

#295 • Image Production

Why can a virtual grid reduce patient dose compared to a physical grid?

Answer

A physical grid requires increased technique to compensate for the radiation it absorbs, while a virtual grid can allow certain exposures without a physical grid at all.

#296 • Image Production

What is a Potter-Bucky diaphragm and why does it move?

Answer

It is a moving grid housed in the Bucky assembly that oscillates or reciprocates during the exposure to blur out the grid lines that a stationary grid would otherwise show.

#297 · Image Production

What is quantization in digital image processing?

Answer

Quantization is the process where the analog-to-digital converter samples the detector's analog signal and converts it into discrete numeric values the computer can work with.

#298 · Image Production

What does flat fielding correct for?

Answer

It corrects for the fact that a detector's response is not perfectly uniform across its surface, since some pixels are naturally more or less sensitive than others.

#299 · Image Production

What is dead pixel correction?

Answer

Dead pixel correction identifies detector elements that have failed and substitutes interpolated values from surrounding healthy pixels so the dead pixel does not appear as a distracting black or white dot on the final image.

#300 · Image Production

What is the histogram used for during image preprocessing?

Answer

The histogram graphs how many pixels in the image fall at each exposure value, and the software analyzes it to identify the anatomy present and calculate an appropriate exposure index.

#301 • Image Production

What does edge enhancement processing do?

Answer

It increases the contrast at boundaries between different tissue densities, making structures like bone edges or catheter tips appear sharper.

#302 • Image Production

What does equalization processing accomplish?

Answer

It compresses the very bright and very dark ends of the histogram so that both dense and lucent areas of a single image, such as the mediastinum and lung fields on a chest image, can be evaluated on the same display.

#303 • Image Production

What does smoothing do to a digital image, and what is the tradeoff?

Answer

Smoothing reduces the visual appearance of noise, sometimes at the cost of fine detail, and is often applied automatically for lower-dose exposures where noise would otherwise be more prominent.

#304 • Image Production

What is the purpose of values of interest (VOI) processing and a look-up table (LUT)?

Answer

They map the wide range of digital pixel values down to the specific brightness levels the display monitor and human eye can distinguish, tailoring the detector's huge dynamic range into an image that looks correctly exposed on screen.

#305 • Image Production

What are window level and window width adjustments?

Answer

Window level adjusts image brightness and window width adjusts image contrast, both applied as postprocessing after the image is generated.

#306 • Image Production

What does electronic cropping or masking do to a digital image?

Answer

It removes unwanted areas from the displayed or printed image, most often the collimated border, without altering the underlying acquired data.

#307 · Image Production

Why does ambient room lighting matter for reading rooms?

Answer

Ambient lighting that is too bright washes out subtle gray-scale differences on the display monitor, which is why reading rooms are kept dim.

#308 · Image Production

Why can't a low-resolution monitor display the fine detail a high-resolution detector captured?

Answer

A monitor's spatial resolution depends on pixel size and pixel pitch, and a monitor with coarser pixels cannot display detail finer than its own pixel structure allows, no matter how good the original acquisition was.

#309 • Image Production

What is DICOM and why does it matter?

Answer

DICOM (Digital Imaging and Communications in Medicine) is the universal standard format that allows equipment from different manufacturers to communicate images and data with each other reliably.

#310 • Image Production

What is PACS?

Answer

The Picture Archiving and Communication System is the network that stores and distributes digital images across a facility.

#311 • Image Production

What is the difference between an HIS/RIS and an EMR/EHR?

Answer

The Hospital Information System and Radiology Information System manage scheduling, patient demographics, orders, and reporting, while the Electronic Medical Record or Electronic Health Record holds the patient's broader clinical history.

#312 • Image Production

Why does every department need documented downtime procedures?

Answer

Because patient care cannot simply stop when PACS, RIS, or the network fails. Downtime procedures cover how images and orders will be tracked, how critical results will be communicated, and how the backlog will be reconciled once systems are restored.

#313 • Image Production

Why does light field to radiation field alignment need to be checked as a QC test?

Answer

This test exists because a collimator that drifts out of alignment can expose tissue the technologist never intended to irradiate, increasing dose for no benefit, or can cut off anatomy the physician needs to see, leading to a repeat exposure.

#314 • Image Production

What does central ray alignment QC verify, and why does it matter?

Answer

It verifies the x-ray beam is perpendicular to and centered on the image receptor. A central ray that is off-center or angled can introduce image distortion and throw off grid alignment, causing grid cutoff.

#315 • Image Production

Why does plate uniformity testing catch problems that would otherwise reach a patient image?

Answer

It exposes the detector to a flat, uniform beam and checks that every part of the resulting image reads the same value, which catches detector damage, debris, or component aging that would otherwise show up as an unexplained artifact on a real patient image.

#316 • Image Production

What does spatial resolution QC testing of a digital detector confirm?

Answer

It uses a test pattern to confirm the detector is still resolving fine detail at the level it was designed for, catching gradual degradation before it affects diagnostic accuracy.

#317 • Image Production

Why do detectors need periodic calibration even though they worked fine when installed?

Answer

Detector elements can drift or degrade with use and radiation exposure over time, so calibration resets the detector's baseline response to keep its sensitivity accurate and consistent over its lifespan.

#318 • Image Production

What is the grayscale standard display function (GSDF) and why does it matter?

Answer

It is a calibration standard ensuring a given change in pixel value produces a perceptually consistent change in displayed brightness across the monitor's entire range, matching how the human eye actually perceives differences in gray.

#319 • Image Production

Why is periodic monitor luminance testing needed even if a monitor looks fine to the eye?

Answer

An aging monitor backlight can quietly reduce contrast resolution well before the change is obvious to the naked eye, so luminance testing confirms maximum brightness and consistency have not degraded.

#320 • Image Production

Why does shielding accessory QC test lead aprons, gloves, and thyroid shields for cracks or holes?

Answer

The lead composite material can develop cracks or holes as it ages and is folded or dropped repeatedly. A cracked apron gives a false sense of protection since staff and patients assume they are shielded when radiation can actually pass through the damaged area.

#321 • Procedures

Lateral skull: CR entry point, and how do you confirm there is no rotation?

Answer

CR is perpendicular to the IR, entering 2 inches (5 cm) superior to the EAM, with no rotation or tilt. A correctly positioned lateral shows superimposed orbital roofs and superimposed mandibular rami; failure of these paired structures to line up is the most common rotation error.

#322 • Procedures

Towne method: CR angle, entry point, what it demonstrates, and how rotation is confirmed?

Answer

CR angles 30 degrees caudad to the OML, entering about 2.5 inches (6 cm) above the glabella and exiting near the foramen magnum, demonstrating the occipital bone, foramen magnum, and dorsum sellae. The dorsum sellae and posterior clinoids should project within the foramen magnum with symmetric petrous ridges confirming no rotation.

#323 · Procedures

Caldwell method: CR angle, exit point, and what it demonstrates? How does it differ from a plain PA skull?

Answer

Forehead and nose touch the IR, CR angles 15 degrees caudad and exits at the nasion, demonstrating the frontal bone, frontal sinuses, and superior orbital margins, with petrous ridges projecting into the lower third of the orbits. A plain PA skull uses the same positioning but a perpendicular CR, and the petrous ridges fill the orbits instead of dropping below them.

#324 · Procedures

Submentovertex (SMV) skull: positioning, CR, and what it demonstrates?

Answer

The neck is hyperextended until the IOML is as parallel to the IR as tolerated, vertex touching the IR, CR perpendicular to the IOML entering along the midsagittal plane about 3/4 inch (2 cm) anterior to the EAMs. It demonstrates the skull base, sphenoid and ethmoid sinuses, and mandible, with the mandibular condyles projecting anterior to the petrous pyramids.

#325 • Procedures

Trauma cross-table lateral skull: how is it performed when the neck cannot be moved?

Answer

The patient stays supine, the IR is placed vertically against the side of the head, and the CR is directed horizontally, perpendicular to the IR, entering 2 inches (5 cm) above the EAM, preserving cervical immobilization.

#326 • Procedures

What is the general rule for adapting skull and facial projections when cervical injury cannot be excluded?

Answer

Never flex, extend, or rotate the neck. Adapt every skull and facial projection to a horizontal-beam or non-rotational technique that keeps the neck stationary.

#327 · Procedures

Waters method: positioning and what it best demonstrates?

Answer

The neck is extended so the chin rests on the IR with the OML at 37 degrees to the IR, CR perpendicular exiting at the acanthion. It is the best all-around facial view for the maxilla, zygomatic arches, and orbital floors.

#328 · Procedures

When is the modified Waters projection used?

Answer

To evaluate the orbital floor for a blowout fracture. It uses the same acanthion-centered Waters positioning, but the OML is angled to 55 degrees instead of the standard 37, which elongates the orbital floor and drops the petrous ridges lower, out of the way of the orbit.

#329 · Procedures

Why is a modified Waters at 55 degrees preferred for evaluating the orbital floor?

Answer

Angling the OML to 55 degrees instead of the standard 37 degrees elongates the orbital floor and drops the petrous ridges lower, out of the way of the orbit.

#330 · Procedures

How does the lateral facial bones CR entry point differ from a standard lateral skull?

Answer

Standard lateral skull positioning is used, but the CR enters lower, at the zygomatic bone near the outer canthus, to center on facial rather than cranial structures.

#331 • Procedures

Lateral nasal bones: where does the CR enter?

Answer

Same positioning as lateral facial bones, tightly collimated to the nasal bones, with the CR entering about half an inch inferior to the nasion.

#332 • Procedures

Axiolateral oblique mandible: positioning and CR?

Answer

The head rotates 30 to 45 degrees toward the side being examined, CR angled 25 degrees cephalad, entering the far mandibular ramus and exiting near the near mandibular angle. This frees the body and ramus from cervical spine and opposite-side superimposition.

#333 · Procedures

PA mandible: CR and what it best shows?

Answer

Forehead and nose contact the IR, CR is perpendicular and exits at the acanthion. It best shows the rami and body of the mandible without rotation.

#334 · Procedures

Which mandible projection is the projection of choice for condylar fracture or dislocation?

Answer

The AP axial Towne for the condyles, using standard Towne positioning (OML perpendicular, CR 30 degrees caudad) but centered lower to profile the condyles within the mandibular fossae.

#335 · Procedures

PA axial mandible: how does it differ from a plain PA mandible?

Answer

Same forehead-nose positioning, but with 20 to 25 degrees of caudal CR angulation added, which better demonstrates the body and symphysis.

#336 · Procedures

TMJ modified Law method: positioning, CR, and why is it done open and closed mouth?

Answer

The head is in a true lateral rotated 15 degrees toward the IR, CR angled 15 degrees caudad, entering about 2 inches (5 cm) superior to the upper EAM. Comparing open- and closed-mouth images shows normal condylar translation out of the fossa on opening; failure to translate suggests TMJ dysfunction.

#337 · Procedures

TMJ modified Towne method: CR angle?

Answer

Standard Towne positioning is used but the CR angles 30 to 35 degrees caudad and is centered on the TMJs, again taken open- and closed-mouth.

#338 · Procedures

What is the general technical rule for imaging paranasal sinuses, and why?

Answer

Sinuses are imaged erect with a horizontal beam, specifically to demonstrate air-fluid levels. If the erect, horizontal-beam technique is not preserved, any fluid line disappears from the image.

#339 · Procedures

Lateral sinus projection: CR entry point?

Answer

Erect, true lateral, CR horizontal, entering about 0.5 inch (1.25 cm) posterior to the outer canthus, showing all four sinus groups in profile.

#340 · Procedures

Which horizontal-beam sinus projection is considered the workhorse view, and why?

Answer

The Waters method with a horizontal beam (chin on IR, OML at 37 degrees) best shows the maxillary sinuses, which is why it is the workhorse sinus projection.

#341 · Procedures

AP axial cervical spine: CR angle and entry point?

Answer

CR angles 15 to 20 degrees cephalad, entering at C4 (thyroid cartilage level), demonstrating C3 through C7 and the disc spaces.

#342 · Procedures

AP open-mouth odontoid: positioning and correct dens appearance?

Answer

The mouth opens fully so the occlusal plane and skull base are superimposed, CR perpendicular through the center of the open mouth. The dens should sit centered between the C1 lateral masses with equal space on each side.

#343 · Procedures

Fuchs method: indication and positioning?

Answer

Used for the AP dens view when the patient cannot hold the mouth open and still. The neck is hyperextended until a line from the mentum to the mastoid tip is perpendicular to the IR, CR perpendicular just below the mandible tip.

#344 · Procedures

Lateral cervical spine: SID, CR, and what must be visible on the image?

Answer

Usually erect at a long 72 inch (180 cm) SID, midsagittal plane parallel to the IR, shoulders relaxed and depressed, CR perpendicular entering at C4. All seven cervical vertebrae plus the C7-T1 junction must be shown, using gentle traction or weighted wrists if the shoulders obscure C7.

#345 · Procedures

Swimmer's lateral: indication and technique?

Answer

Used to show the cervicothoracic junction (C7-T1) when the shoulders obscure it on a routine lateral. The arm nearer the IR raises overhead while the opposite shoulder depresses, CR horizontal, angled 3 to 5 degrees caudad, entering at T1.

#346 · Procedures

PA oblique versus AP oblique cervical spine: which shows which foramina?

Answer

PA obliques best show the intervertebral foramina and pedicles closest to the IR, while AP obliques show those farthest from the IR. Both are taken at 45 degrees rotation, right and left.

#347 · Procedures

When are cervical flexion and extension laterals performed, and when are they contraindicated?

Answer

They assess ligamentous stability with the chin tucked (flexion) or tilted back (extension), CR perpendicular at C4. They are never done in acute trauma until fracture is ruled out, since moving an unstable spine risks cord injury.

#348 · Procedures

AP thoracic spine: breathing technique and CR?

Answer

Shallow, continuous breathing with a long exposure blurs ribs and lung markings while keeping the spine sharp. CR is perpendicular, entering about 3 to 4 inches (7 to 10 cm) below the jugular notch at T7.

#349 · Procedures

Lateral thoracic spine: what are the two breathing approaches used, and why?

Answer

Shallow breathing blurs the lungs for the standard view, while full expiration is used for the lower thoracic vertebrae, since expiration elevates the diaphragm and reduces density over T10-T12.

#350 · Procedures

What is the purpose and technique of the scoliosis series?

Answer

It measures spinal curvature for Cobb angle assessment, taken erect on a single long IR from C7 to the sacrum, PA or lateral, with no compression band and no angulation so the spine's natural alignment is preserved.

#351 · Procedures

AP lumbar spine: positioning and CR?

Answer

Supine, knees flexed to flatten the lordotic curve, CR perpendicular entering at the iliac crest level (L4).

#352 · Procedures

Lateral lumbar spine: positioning and CR?

Answer

True lateral, knees and hips flexed, a support placed under the waist to keep the spine parallel to the tabletop, CR perpendicular at the iliac crest (L4).

#353 • Procedures

Why is a dedicated L5-S1 lateral spot view needed, and what is the CR angle?

Answer

The wings of the ilium often obscure L5-S1 on the routine lateral. A tightly collimated CR angles 5 to 8 degrees caudad, entering about 2 inches (5 cm) inferior and posterior to the iliac crest.

#354 • Procedures

What do the lumbar oblique projections show, and which structures are best demonstrated closest to the IR?

Answer

They show the 'Scotty dog' appearance of the pars interarticularis, used to detect spondylolysis. On an anterior oblique (prone), structures closer to the IR are best shown; CR enters about 2 inches (5 cm) medial to the elevated side at L3.

#355 • Procedures

AP axial L5-S1 projection: CR angle for males versus females?

Answer

Supine, CR angled 30 to 35 degrees cephalad for males or 35 to 40 degrees for females to compensate for a typically greater pelvic curve, entering midline superior to the pubic symphysis.

#356 • Procedures

Sacrum and coccyx: why are they imaged as separate projections, and what are the CR angles?

Answer

Their curvature would foreshorten both on one straight AP. The AP axial sacrum angles the CR 15 degrees cephalad, entering about 2 inches (5 cm) superior to the pubic symphysis; the AP axial coccyx uses the opposite angle, 10 degrees caudad, from the same entry point.

#357 · Procedures

Lateral sacrum and coccyx: CR entry point?

Answer

True lateral positioning, CR perpendicular entering about 3.5 inches (9 cm) posterior to the ASIS, roughly at the level of the greater trochanter.

#358 · Procedures

AP axial sacroiliac (SI) joints: CR angle and entry?

Answer

Supine, CR angled 30 to 35 degrees cephalad, entering midline about 2 inches (5 cm) superior to the pubic symphysis, opening the naturally angled joint spaces.

#359 · Procedures

Posterior oblique versus anterior oblique SI joint projections: which joint shows farther from the IR?

Answer

Posterior obliques (25 to 30 degrees from supine) show the SI joint farther from the table, CR entering about 1 inch (2.5 cm) medial to the elevated ASIS. Anterior obliques show the joint closer to the IR.

#360 · Procedures

AP pelvis: leg positioning, why, and how does a correctly positioned image look?

Answer

Legs are internally rotated 15 to 20 degrees to correct for normal femoral anteversion and profile the femoral necks, CR perpendicular entering midline midway between the ASIS and pubic symphysis. A correct image shows symmetric iliac wings and obturator foramina, with the lesser trochanters minimal or absent since internal rotation tucks them behind the femoral shaft.

#361 · Procedures

AP hip (unilateral): CR entry point?

Answer

Same internal leg rotation as the AP pelvis, centered and collimated to one hip. CR is perpendicular, entering about 1 inch (2.5 cm) inferior to a line from the ASIS to the pubic symphysis, at the femoral neck level.

#362 · Procedures

Unilateral frog-leg lateral hip: what is it especially good for, and how is it positioned?

Answer

It is especially good for subtle femoral neck fractures or slipped capital femoral epiphysis in non-trauma patients. The hip and knee flex and the thigh abducts so the foot sole rests against the opposite knee, CR perpendicular at the femoral neck.

#363 · Procedures

Danelius-Miller method: what is it and how is it performed?

Answer

It is the cross-table, horizontal-beam lateral hip used for suspected hip fracture. The unaffected leg is flexed and lifted out of the beam's path, the IR sits vertically against the lateral hip, and CR is directed horizontally, perpendicular to the femoral neck, entering the groin crease on the affected side, all without moving the injured leg.

#364 · Procedures

Clements-Nakayama method: when is it used, and what is the CR angle?

Answer

Used when even the unaffected leg cannot be raised, or when both hips may be injured. The CR angles 15 to 20 degrees from horizontal, directed from below and lateral to the affected hip, entering at the femoral neck, without moving the injured leg.

#365 • Procedures

Bilateral frog-leg pelvis: how is it positioned, and when is it commonly used?

Answer

Both legs are abducted, externally rotated, and knees flexed, comparing both hips at once. It is common in pediatric hip screening.

#366 • Procedures

Inlet versus outlet AP axial pelvis views: CR angles and what each assesses?

Answer

The inlet view angles the CR 40 degrees caudad, entering midline about 2 inches (5 cm) below the ASIS level, assessing anteroposterior displacement of a pelvic ring injury. The outlet view angles the CR 35 to 40 degrees cephalad, entering midline at the pubic symphysis, assessing superior-inferior displacement.

#367 · Procedures

Judet method: what is it, and what does it evaluate?

Answer

It is a posterior oblique acetabulum projection, rotating the patient 45 degrees posteriorly (both right and left are obtained), CR perpendicular entering about 2 inches (5 cm) inferior and 2 inches medial to the elevated ASIS. It evaluates the acetabular rims and columns after dislocation or fracture.

#368 · Procedures

What is the purpose of an AP pelvis weight-bearing view?

Answer

It assesses pelvic ring stability and symphysis widening under physiologic load with the patient standing, an instability that a supine image cannot show.

#369 • Procedures

What is the unifying trauma-imaging principle that explains cross-table laterals, the swimmer's lateral, and the Clements-Nakayama method?

Answer

Never move an injured part more than absolutely necessary. A horizontal beam or angled CR is used so the tube and IR do the work of achieving a diagnostic image instead of manipulating the patient.

#370 • Procedures

On an AP or PA skull, spine, or pelvis image, what visual signature indicates rotation?

Answer

Asymmetric distance from the spinous processes to the pedicles on a spine image, or asymmetric obturator foramina and iliac wings on a pelvis image, both indicate rotation and require repositioning.

#371 · Procedures

Where does the central ray enter for a PA chest radiograph?

Answer

Perpendicular to the midsagittal plane at the level of T7, roughly the inferior scapular angle.

#372 · Procedures

Why does a PA projection beat an AP projection for chest imaging?

Answer

The heart sits anteriorly, so PA positioning keeps it closer to the receptor for a more accurate size, and standing upright lets the diaphragm drop so free fluid or free air settles where it is easiest to see.

#373 · Procedures

What breathing instruction and rib count apply to a PA chest radiograph?

Answer

Full inspiration, with about ten posterior ribs visible above the diaphragm on a well-inflated image.

#374 · Procedures

When is an AP chest projection used instead of PA?

Answer

When the patient cannot manage a PA, most often trauma or ICU patients; it is taken AP upright or AP supine, and the heart appears magnified in both.

#375 · Procedures

Why is the left side placed against the receptor for a lateral chest?

Answer

The heart is left sided, so keeping the left side against the receptor keeps the heart closest to it and minimizes magnification.

#376 · Procedures

What is the central ray placement and breathing instruction for a lateral chest?

Answer

Horizontal and perpendicular at T7 at the mid-coronal plane, taken on full inspiration with the arms raised overhead to clear the soft tissue.

#377 · Procedures

What does the AP lordotic chest projection accomplish?

Answer

It throws the clavicles up and off the lung apices so the apices can be evaluated free of clavicular overlap.

#378 · Procedures

How is the patient positioned for an AP lordotic chest projection?

Answer

Standing about a foot from the receptor and leaning back so the shoulders touch it while the body angles away; the central ray stays perpendicular to the receptor at T7.

#379 · Procedures

For lateral decubitus chest positioning, which side goes down for a suspected pleural effusion and which side goes up for a suspected pneumothorax?

Answer

The affected side goes down for an effusion, since fluid pools to the dependent side, and the affected side goes up for a pneumothorax, since air rises.

#380 · Procedures

How is rotation identified on a PA chest radiograph?

Answer

By comparing the distance from each sternoclavicular joint to the vertebral spinous process; unequal distances mean the patient was turned, which distorts heart size and mediastinal contours.

#381 · Procedures

What breathing instruction is used for ribs above the diaphragm and why?

Answer

Full inspiration, positioned like a chest, because the air-filled lung provides the contrast needed to see the ribs clearly.

#382 · Procedures

What breathing instruction is used for ribs below the diaphragm and why?

Answer

Full expiration, because raising the diaphragm out of the way keeps it from overlapping the lower ribs.

#383 · Procedures

What is the purpose of rib oblique positioning?

Answer

RAO or LAO prone, or RPO or LPO supine, at about 45 degrees throws the axillary portion of the ribs into profile, clear of superimposition.

#384 · Procedures

Why is the sternum imaged RAO instead of straight PA?

Answer

A straight PA sternum overlies the spine and heart; rotating 15 to 20 degrees from prone throws the sternum into the homogeneous heart shadow where it can be seen clearly.

#385 · Procedures

What breathing technique is used for the RAO sternum and why?

Answer

Shallow breathing continued throughout the exposure, not suspended, which blurs the overlying ribs while the more rigid sternum stays sharp.

#386 · Procedures

What must a lateral sternum radiograph show?

Answer

The whole sternum in profile without rib overlap, taken true lateral on full inspiration with the arms pulled back.

#387 • Procedures

What is the central ray level for soft tissue neck radiographs, and why should the patient avoid phonating?

Answer

Perpendicular at the thyroid cartilage, around C6-C7; making a sound changes the airway's shape and position and would blur the structures being evaluated.

#388 • Procedures

Why does a PA projection of the SC joints only work as a scout, and how is one joint isolated?

Answer

The SC joints sit over the spine on a PA, so it only shows both joints together; the RAO isolates the right joint and the LAO isolates the left, each with a 10 to 15 degree rotation.

#389 · Procedures

What does the AP or PA axial clavicle projection add beyond the standard AP or PA clavicle?

Answer

Angling the central ray 15 to 30 degrees cephalad for AP or caudad for PA straightens the clavicle's natural curve and clears it of the ribs and scapula, since the straight AP or PA foreshortens that curve.

#390 · Procedures

For PICC tip confirmation, where is the central ray directed and where must the tip terminate?

Answer

Central ray perpendicular at T7 on a portable AP chest, and the tip must terminate at the cavoatrial junction, where the superior vena cava meets the right atrium.

#391 · Procedures

What is the central ray location, level, and breathing instruction for a KUB?

Answer

Perpendicular at the iliac crest, centering roughly L4-L5, exposed on expiration to reduce breathing motion.

#392 · Procedures

What anatomy must be included on a KUB and why?

Answer

The diaphragm and the symphysis pubis both, since missing either edge could mean a free air or pelvic finding is missed.

#393 · Procedures

Where is the central ray placed on an AP upright abdomen and why?

Answer

About two inches above the iliac crest to keep the diaphragm in the field, since both hemidiaphragms must be visible to catch free air.

#394 · Procedures

Why is the patient positioned on the left side for a left lateral decubitus abdomen?

Answer

Free air rises over the liver on the left side down, where it shows up clearly against solid tissue rather than against the gas-filled stomach.

#395 · Procedures

When is a dorsal decubitus abdomen used?

Answer

When the patient cannot lie on either side; it is taken supine with a horizontal beam from the side, central ray at the iliac crest along the midcoronal plane.

#396 · Procedures

Why is the RAO the workhorse projection for an esophagram?

Answer

Rotating 35 to 40 degrees from prone clears the esophagus off the spine and heart shadow, with the central ray perpendicular at T5-T6.

#397 · Procedures

What does the left lateral esophagram projection show?

Answer

The retrocardiac esophagus and any posterior indentation from an enlarged left atrium, at the same T5-T6 level.

#398 · Procedures

How is a modified barium swallow performed?

Answer

Under live fluoroscopy with varied food and liquid consistencies, so positioning is directed moment to moment rather than following a fixed protocol.

#399 · Procedures

Why must the patient be NPO at least eight hours before an upper GI series?

Answer

So the stomach is empty, since retained food dilutes contrast and can mimic pathology.

#400 · Procedures

Why are patients asked to avoid smoking and gum chewing before an upper GI series?

Answer

Both stimulate gastric secretions and peristalsis that can wash barium off the mucosa before the images are taken.

#401 · Procedures

What does the RAO projection demonstrate on an upper GI series?

Answer

The duodenal bulb and loop free of overlap, rotating 40 to 70 degrees from prone with the central ray at L1-L2.

#402 · Procedures

What does the PA projection show on an upper GI series?

Answer

The stomach body, taken straight prone at L1-L2 on suspended expiration.

#403 · Procedures

What does the right lateral projection show on an upper GI series?

Answer

The retrogastric space and duodenal bulb in profile, right side against the receptor at L1-L2.

#404 · Procedures

Why is the LPO used on an upper GI series?

Answer

Barium sinks to the fundus while gas rises to the antrum and duodenal bulb as the patient is rotated left posterior, filling the fundus with barium, which is useful for double contrast studies.

#405 · Procedures

What is the most common positioning error on the upper GI RAO and LPO obliques?

Answer

Under-rotating the body, which leaves the duodenal bulb or fundus still superimposed over the spine or antrum instead of clear of it.

#406 · Procedures

How does a small bowel follow-through track barium?

Answer

A PA scout at L2 on suspended expiration is followed by PA images at timed intervals, commonly every 15 to 30 minutes, tracking the barium column through the jejunum and ileum.

#407 · Procedures

Why are spot images with compression taken once contrast reaches the ileocecal valve?

Answer

To demonstrate the terminal ileum clearly, an area where Crohn disease findings are commonly sought.

#408 · Procedures

Why does a contrast enema require thorough bowel prep beforehand?

Answer

So retained stool does not mimic a lesion and contrast coats the mucosa evenly.

#409 · Procedures

What does the left lateral rectum projection evaluate on a contrast enema?

Answer

The rectosigmoid, taken before the rest of the colon fills, central ray perpendicular at the level of the ASIS, centered to the mid-coronal plane posterior to the ASIS, between the ASIS and the posterior sacrum.

#410 · Procedures

What do the lateral decubitus views show on a contrast enema?

Answer

The medial and lateral colon walls, using the air-barium interface with a horizontal beam at the iliac crest.

#411 · Procedures

Why are LPO and RPO used on a contrast enema?

Answer

To open the splenic and hepatic flexures, since they overlap each other on a straight AP.

#412 · Procedures

What does the PA projection show on a contrast enema?

Answer

The transverse colon and both flexures together, central ray at the iliac crest.

#413 · Procedures

What is the purpose of the axial sigmoid projection on a contrast enema, and what angle is used?

Answer

It unfolds the sigmoid loop, which otherwise hides polyps by overlapping itself, using a central ray angled 30 to 40 degrees caudad for PA or cephalad for AP.

#414 · Procedures

What does a post-evacuation view on a contrast enema check?

Answer

Mucosal detail and how completely the colon empties, central ray perpendicular at the iliac crest.

#415 • Procedures

What is the most common error during the double contrast air step of a contrast enema?

Answer

Losing rectal tube retention or over-insufflating, which either drains contrast prematurely and underfills the colon or distends it enough to flatten and hide polyps.

#416 • Procedures

How is the patient positioned for surgical cholangiography, and what breathing instruction is given?

Answer

Positioning follows the surgical field, often a slight RAO to clear the duct off the spine; since the patient is under anesthesia, the anesthesiologist may briefly suspend ventilation instead of a direct breathing instruction.

#417 · Procedures

How is the patient positioned during ERCP, and what is the radiographer's role?

Answer

Typically prone or semi-prone for endoscope access; the radiographer operates the fluoroscopic unit and takes spot images as directed by the gastroenterologist.

#418 · Procedures

What anatomic coverage is required for a foreign body ingestion series?

Answer

Enough anatomy from the pharynx to the rectum, since the object can move well beyond where it was last seen.

#419 · Procedures

What is the gold standard method for confirming feeding tube placement?

Answer

A portable AP chest or abdomen radiograph, which directly shows whether the tip has crossed the diaphragm into the stomach rather than coiling in the esophagus or entering the airway.

#420 · Procedures

Why is breath suspended on every exposure during cystography?

Answer

Because motion blurs the bladder outline.

#421 · Procedures

Why is the female voiding cystourethrogram taken AP?

Answer

The short, roughly 4-centimeter female urethra runs nearly straight down in the midline, so an AP view during voiding shows its full length without unfolding any curve.

#422 · Procedures

Why is the male voiding cystourethrogram taken RPO instead of AP?

Answer

The longer male urethra curves through the prostatic and penile segments, and a 30 degree posterior oblique elongates and unfolds that curve so a stricture anywhere along it can be seen without foreshortening.

#423 · Procedures

Why is breathing suspended on expiration for each exposure during retrograde urography?

Answer

Because the kidneys and ureters move with respiration.

#424 · Procedures

When in the menstrual cycle is a hysterosalpingogram scheduled, and why?

Answer

In the first ten days of the cycle, after bleeding stops but before ovulation, to avoid an early pregnancy and get the clearest cavity image.

#425 · Procedures

What does the AP scout accomplish before contrast is instilled for cystography?

Answer

It confirms there is no residual contrast in the bladder before filling, central ray perpendicular about two inches above the symphysis pubis.

#426 · Procedures

PA hand: CR location and what the image must demonstrate

Answer

CR is perpendicular to the third metacarpophalangeal joint. The image must show the distal forearm through all phalanges with open joint spaces.

#427 · Procedures

What is the fan lateral hand position and why is it used?

Answer

The fingers are spread like a fan so each digit clears its neighbor on the lateral projection. It is useful for showing anterior or posterior displacement of fracture fragments.

#428 · Procedures

Why is the PA oblique hand rotated 45 degrees instead of taken flat?

Answer

Rotating 45 degrees with the fingers supported parallel to the IR opens the metacarpal spaces that overlap and superimpose on a straight PA view.

#429 · Procedures

Isolated PA finger: CR location and technique

Answer

CR centers on the proximal interphalangeal joint of the affected digit, with the adjacent fingers flexed out of the way.

#430 · Procedures

Why is a true lateral finger necessary in addition to the PA finger?

Answer

The lateral finger is critical for showing anterior or posterior displacement of a fracture fragment, which a PA projection would hide.

#431 · Procedures

How is the AP thumb obtained?

Answer

The whole hand is rotated medially until the posterior surface of the thumb rests on the IR, with CR at the first MCP joint, giving a true AP of the digit.

#432 · Procedures

How are the lateral and oblique thumb positions obtained?

Answer

The routine pronated PA hand position gives an oblique thumb for free, roughly 45 degrees from lateral, since the thumb's natural resting rotation relative to the fingers already places it obliquely. A true lateral thumb instead requires additional rotation beyond the routine PA hand position.

#433 · Procedures

What is the purpose of ulnar deviation on a PA wrist projection?

Answer

In neutral position the scaphoid is foreshortened and can hide a fracture line. Deviating the wrist toward the ulna pulls the scaphoid away from the other carpals and elongates it, correcting the foreshortening.

#434 · Procedures

Why is the scaphoid clinically significant among the carpal bones?

Answer

The scaphoid is the carpal bone most often fractured and is most prone to avascular necrosis if a fracture is missed.

#435 · Procedures

What is the Stecher method and why is it used?

Answer

The Stecher method is a PA axial scaphoid projection that adds a 20 to 25 degree proximal CR angle with the wrist still in ulnar deviation, elongating the scaphoid even further than ulnar deviation alone to clear a fracture line.

#436 · Procedures

What is the Gaynor-Hart method and what does it demonstrate?

Answer

The Gaynor-Hart method is a tangential carpal canal projection that hyperextends the wrist palm up with CR angled 25 to 30 degrees along the hand's long axis into the carpal canal. It demonstrates the pisiform, the hamulus of the hamate, and the tunnel that carpal tunnel syndrome affects.

#437 · Procedures

What is a bone age study and what does it use for comparison?

Answer

A single PA of the left hand and wrist is compared against a standardized reference atlas, classically Greulich and Pyle, to estimate skeletal maturity. It is useful for evaluating growth disorders or predicting adult height.

#438 · Procedures

PA, lateral, and oblique wrist: CR location

Answer

CR is centered at the midcarpal area for the standard PA, lateral, and oblique wrist projections.

#439 · Procedures

Forearm AP: positioning and why

Answer

The arm is fully extended with the hand supinated so the radius and ulna lie parallel without crossing, CR at the mid-forearm, with both the wrist and elbow joints included.

#440 · Procedures

Why does a true lateral forearm require the thumb-up position?

Answer

The elbow is flexed 90 degrees with the hand thumb-up because that position keeps the radius and ulna from overlapping, which is required for a true lateral.

#441 • Procedures

Elbow AP: positioning and what it demonstrates

Answer

The arm is extended with the hand supinated, CR perpendicular to the joint, demonstrating the joint space free of superimposition along with the proximal radius, ulna, and distal humerus.

#442 • Procedures

Lateral elbow: flexion angle and CR

Answer

The elbow is flexed exactly 90 degrees with the hand thumb-up, CR directed at the joint.

#443 · Procedures

On a correctly positioned lateral elbow, what do the anterior and posterior fat pads indicate?

Answer

The anterior fat pad is normally a faint visible sliver. The posterior fat pad is normally invisible, and its appearance signals a joint effusion, often from an occult fracture.

#444 · Procedures

Medial oblique elbow: rotation and what it best shows

Answer

The arm is rotated about 45 degrees medially, best demonstrating the coronoid process.

#445 · Procedures

Lateral oblique elbow: rotation and what it best shows

Answer

The arm is rotated about 45 degrees laterally, best demonstrating the radial head, neck, and capitulum.

#446 · Procedures

When is the AP partial flexion technique used at the elbow, and how is it performed?

Answer

It is used when a patient cannot fully extend the elbow. Two exposures are taken, one centered on the distal humerus and one on the proximal forearm, each with CR entering wherever that segment sits closest to the IR, avoiding forced extension that a possible fracture makes unsafe.

#447 · Procedures

What is the Coyle method and when is it used?

Answer

The Coyle method uses trauma axial laterals of the elbow when even partial flexion is not tolerated. With the elbow flexed near 90 degrees, it works around whatever limited motion the elbow allows.

#448 · Procedures

Coyle method: what do the two CR angles each demonstrate?

Answer

One exposure angles CR 45 degrees toward the shoulder, showing the radial head, neck, and capitulum, useful for radial head fractures. The second angles CR 45 degrees toward the hand, showing the coronoid process and trochlea, useful for coronoid process fractures.

#449 · Procedures

Why is the humerus AP taken in neutral rotation when a fracture is suspected?

Answer

Forcing rotation of the arm risks displacing fracture fragments, so the epicondyles are kept parallel to the IR without rotating the arm.

#450 · Procedures

What is the transthoracic lateral humerus and when is it used?

Answer

It is used for a trauma patient instead of a rotational lateral. The injured arm hangs at the side, the other arm raises overhead, and CR passes horizontally through the thorax to project a true lateral without moving the injured arm.

#451 · Procedures

Why does the transthoracic lateral humerus use a breathing technique?

Answer

Shallow breathing during a longer low-mA exposure blurs the ribs and lungs while the denser humerus stays comparatively sharp, since it is one of the few extremity views where breathing technique matters.

#452 · Procedures

AP shoulder internal rotation vs external rotation: what does each profile?

Answer

Internal rotation turns the palm posteriorly, profiling the lesser tuberosity medially. External rotation turns the palm anteriorly, profiling the greater tuberosity laterally and giving the clearest view of the humeral head.

#453 · Procedures

Why is arm rotation avoided when a shoulder fracture or dislocation is suspected?

Answer

Rotating the arm can worsen the injury, which is exactly why the AP neutral rotation and other trauma-specific shoulder views exist.

#454 · Procedures

What is the AP neutral rotation shoulder used for?

Answer

It leaves the arm exactly as the patient presents it and serves as the safe first trauma view when fracture or dislocation is suspected.

#455 · Procedures

What is the Grashey method and why is it the standard for suspected dislocation?

Answer

The Grashey method is a posterior oblique shoulder rotated 35 to 45 degrees from a true AP with the affected side against the IR. Rotating the scapula's flat surface parallel to the IR aligns the beam tangent to the glenohumeral joint, opening the joint space cleanly, which is why it is standard for suspected dislocation or joint narrowing.

#456 · Procedures

Why does a straight AP shoulder fail to show the glenohumeral joint space clearly?

Answer

The glenoid fossa faces anterolaterally rather than straight forward, so a straight AP beam catches the glenoid at a slant and causes it to overlap the humeral head.

#457 · Procedures

What is the Lawrence method and what does it demonstrate?

Answer

The Lawrence method is an inferosuperior axial projection showing the humeral head-glenoid relationship from below, useful for posterior dislocation. The arm abducts 90 degrees if possible, with CR angled 25 to 30 degrees medially through the axilla.

#458 · Procedures

What is the scapular Y view and why is it valuable in trauma?

Answer

The patient is positioned at roughly a 60 degree anterior oblique with the affected shoulder against the IR, so the scapular body, coracoid, and acromion form a Y with the humeral head normally centered at the junction. Since no arm movement is needed, it is valuable in trauma.

#459 · Procedures

How does the scapular Y view distinguish anterior from posterior shoulder dislocation?

Answer

The humeral head is displaced toward whichever arm of the Y corresponds to the direction of dislocation, rather than sitting centered at the junction.

#460 · Procedures

What is the Neer method and what does it evaluate?

Answer

The Neer method is a supraspinatus outlet view, a steeper version of the scapular Y angled 10 to 15 degrees caudad, profiling the outlet space for evaluating impingement.

#461 · Procedures

AP and lateral scapula: positioning and CR

Answer

The AP scapula abducts the arm 90 degrees to draw the scapula off the ribs, CR at the mid-scapular body, sometimes with a breathing technique to blur lung markings. The lateral scapula uses the same oblique body rotation as the scapular Y view.

#462 · Procedures

How are AC joints imaged and why are two exposures taken?

Answer

Both AC joints are imaged bilaterally on one long IR, AP, CR at the midline at joint level. Two exposures are taken, one without weights and one with weights suspended, to reveal a separation the unstressed view might miss.

#463 · Procedures

Why are AC joint stress weights suspended rather than held by the patient?

Answer

Gripping a weight recruits shoulder muscles, which can mask a separation, so the weight is suspended from the wrist instead.

#464 · Procedures

AP entire forefoot: CR location and purpose

Answer

CR centers at the third metatarsophalangeal joint, showing all five toes and distal metatarsals as a screening view.

#465 · Procedures

Individual AP or AP axial toe: CR location

Answer

CR centers on that digit's interphalangeal joint (the sole IP joint on the two-phalanx great toe, or the proximal interphalangeal joint on toes two through five), sometimes angled slightly to open the joint on a short great toe.

#466 · Procedures

Oblique toe and lateral toe: how are they obtained and why is the lateral needed?

Answer

The oblique toe rotates the foot 30 to 45 degrees medially. The lateral toe isolates the affected digit with the other toes taped away, needed to show displacement an AP view cannot.

#467 · Procedures

AP axial foot: CR angle and why it is angled

Answer

CR angles 10 degrees toward the heel, entering the base of the third metatarsal, compensating for the foot's natural arch so the tarsometatarsal joints are not foreshortened.

#468 · Procedures

Medial oblique foot: rotation and clinical role

Answer

The foot rotates 30 degrees medially, opening the sinus tarsi and separating the metatarsal bases. It is the standard oblique for foot trauma.

#469 · Procedures

Lateral foot: CR location

Answer

CR is directed at the base of the third metatarsal on a true mediolateral projection, showing the arch profile.

#470 · Procedures

Why are weight-bearing foot views ordered?

Answer

They are taken standing for arch collapse, flatfoot, or hardware alignment under physiologic load, since a supine image cannot show how the arch behaves under body weight.

#471 · Procedures

Lateral calcaneus: CR location

Answer

CR enters about one inch inferior to the medial malleolus on a straightforward mediolateral projection.

#472 · Procedures

Plantodorsal axial calcaneus: CR angle and why it is so steep

Answer

CR angles 40 degrees cephalad into the plantar surface of the heel. The steep angle is needed to project the calcaneal body clear of the talus and lower leg above it.

#473 · Procedures

AP ankle: positioning and its main limitation

Answer

The leg is extended with the foot dorsiflexed, CR perpendicular midway between the malleoli, showing the distal tibia, fibula, and talus, though the joint space is not fully open because the fibula overlaps the talus slightly.

#474 · Procedures

Mortise ankle view: how much rotation and why is it needed?

Answer

The foot and leg are internally rotated 15 to 20 degrees. The two malleoli are not in the same coronal plane, so this rotation brings the intermalleolar line parallel to the IR and places CR perpendicular to the actual joint.

#475 · Procedures

What does a correctly positioned mortise view show?

Answer

It shows an even joint space around all three sides of the talus with no fibula-tibia overlap, and any narrowing or widening signals ligament injury or fracture that a plain AP would miss.

#476 · Procedures

Lateral ankle: CR location and use

Answer

CR is directed at the medial malleolus on a mediolateral projection, showing the tibiotalar joint from the side and useful for displacement or effusion.

#477 · Procedures

Medial oblique ankle: how does it differ from the mortise view?

Answer

The medial oblique ankle rotates 45 degrees medially, profiling the distal tibiofibular syndesmosis, a larger rotation than the mortise view's 15 to 20 degrees. Both rotate the same direction but serve different purposes at different angles.

#478 · Procedures

AP stress views of the ankle: what do they evaluate?

Answer

Manual inversion or eversion is applied during exposure to evaluate ligament laxity against a normal side.

#479 · Procedures

Why are weight-bearing AP and lateral ankle views ordered?

Answer

They are taken standing to evaluate alignment and joint narrowing under load, often for chronic instability or syndesmotic injury.

#480 · Procedures

Tibia-fibula AP and lateral: why are both adjacent joints included?

Answer

Both the knee and ankle are included with CR at the mid-shaft, since shaft fractures can accompany injury at either adjacent joint.

#481 · Procedures

AP knee: CR location and when is it angled cephalad?

Answer

CR is perpendicular, entering about half an inch below the patellar apex, or angled 3 to 5 degrees cephalad in a larger body habitus, since greater lordotic tilt otherwise foreshortens the joint space.

#482 · Procedures

Why are weight-bearing AP knee views ordered?

Answer

Standing on both legs evaluates joint space narrowing from osteoarthritis, since cartilage loss only shows under load and a supine image can look deceptively normal.

#483 · Procedures

Lateral knee: flexion angle and CR direction

Answer

The knee flexes 20 to 30 degrees, CR angled 5 to 7 degrees cephalad, directed below the medial epicondyle.

#484 · Procedures

Why is the lateral knee not flexed further than 20 to 30 degrees?

Answer

Flexing further tightens the joint and can hide a suprapatellar effusion, a key finding this view is meant to catch.

#485 • Procedures

Medial and lateral knee obliques: what do they clear and evaluate?

Answer

Rotating 45 degrees each way clears the femoral condyles and tibial plateau of overlap and evaluates the fibular head.

#486 • Procedures

What is the general purpose of the intercondylar fossa (tunnel) views of the knee?

Answer

They show the notch between the femoral condyles for osteochondral defects and loose bodies.

#487 · Procedures

What is the Holmblad method?

Answer

The Holmblad method is a PA axial intercondylar fossa projection with the patient kneeling, knee flexed about 70 degrees, CR perpendicular to the lower leg, which because of that steep flexion enters the fossa nearly straight through.

#488 · Procedures

What is the Camp Coventry method and when is it chosen?

Answer

The Camp Coventry method is a PA axial intercondylar fossa projection done prone, knee flexed 40 to 50 degrees, CR angled 40 to 50 degrees caudad to match the flexion. It is chosen when kneeling is not tolerated.

#489 · Procedures

What is the B  cl  re method?

Answer

The B  cl  re method is the supine alternative for the intercondylar fossa, knee flexed 40 to 45 degrees on a sponge, CR angled 40 to 45 degrees cephalad to match, so the beam travels straight through the notch regardless of patient position.

#490 · Procedures

What is the Merchant method and why are both knees imaged together?

Answer

The Merchant method is a tangential patellofemoral projection taken supine with both knees flexed 40 degrees off the table end, CR angled 30 degrees from the long axis of the legs. Both knees are imaged together for side-to-side comparison of subluxation or tracking.

#491 · Procedures

What is the Settegast method and why is it used less often today?

Answer

The Settegast method is a tangential patellofemoral projection taken prone with the knee flexed as sharply as tolerated, CR angled 15 to 20 degrees cephalad into the joint. It is used less often since forced sharp flexion can displace a suspected patellar fracture.

#492 · Procedures

Why is the standalone PA patella taken prone?

Answer

The prone position brings the patella closer to the IR, with CR perpendicular exiting the patella, improving detail.

#493 · Procedures

Femur AP and lateral: which joint is included?

Answer

One adjacent joint, hip or knee, is generally included depending on where the pathology is suspected, with the opposite joint imaged separately if the femur is too long for one IR.

#494 · Procedures

What is a scanogram and how is it performed?

Answer

A scanogram, or long bone measurement, evaluates limb length discrepancy. A radiopaque ruler is placed alongside the limb, and three separate exposures are taken on one long IR, each centered individually over the hip, knee, and ankle with CR perpendicular to that joint.

#495 · Procedures

Why does a scanogram use three separate centered exposures instead of one exposure down the whole leg?

Answer

Centering each joint separately avoids the magnification distortion a single divergent beam would cause, since a joint far from CR would appear artificially larger or smaller than one centered directly under it, corrupting the measurement.

#496 · Procedures

When is a skeletal survey ordered for metastatic disease screening?

Answer

It is ordered in adults since cancers such as breast, prostate, and lung commonly spread to bone, and a survey of the skull, spine, ribs, pelvis, and long bones can catch lytic or sclerotic lesions before a pathologic fracture occurs.

#497 · Procedures

What is a child abuse skeletal survey and why does it image each extremity individually?

Answer

It is a suspected non-accidental trauma workup in a young child. Inflicted injuries often include fractures at different healing stages that are not clinically obvious, so a full survey imaging the skull, chest, spine, pelvis, and each extremity individually documents every skeletal injury present, including old healed fractures the history omits.

#498 · Procedures

What is arthrography and what does it show that plain radiographs cannot?

Answer

Arthrography is a fluoroscopically or ultrasound-guided procedure injecting contrast, either iodinated contrast alone or iodinated contrast plus air, directly into a joint capsule to outline soft tissue such as the capsule lining, cartilage, ligaments, and the menisci or labrum.

#499 · Procedures

When is direct MR arthrography still used despite MRI largely replacing conventional arthrography?

Answer

Direct MR arthrography, injecting dilute gadolinium into the joint before scanning, is still used when the extra distension improves visualization of a subtle labral or ligament tear.

#500 · Procedures

What is the difference between joint aspiration and joint injection?

Answer

Joint aspiration withdraws fluid for diagnostic testing such as infection or gout, or for therapeutic relief of a painful effusion. Joint injection instead delivers a corticosteroid and local anesthetic to reduce inflammation and pain, typically guided by fluoroscopy or ultrasound.